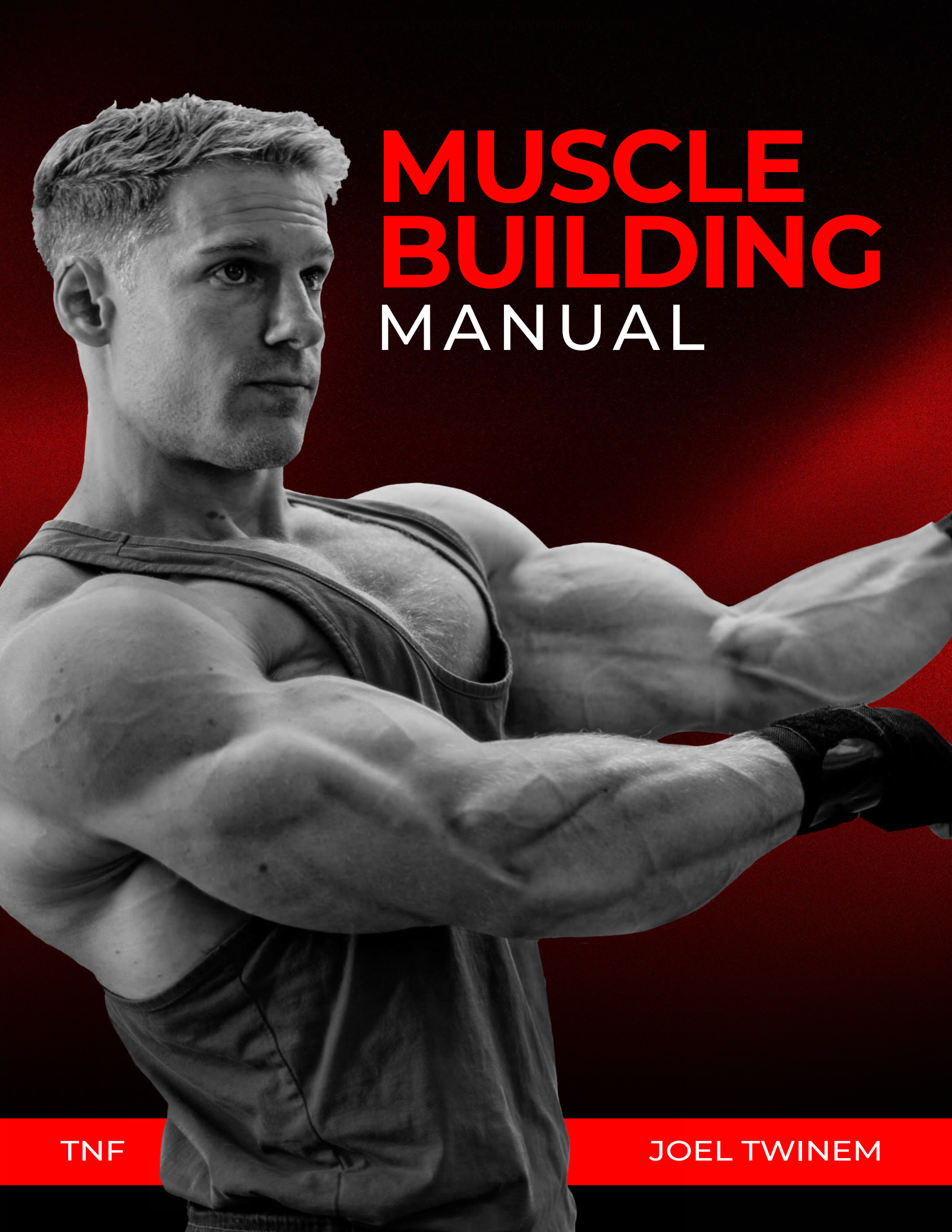




MUSCLE BUILDING MANUAL

TNF

JOEL TWINEM

INTRODUCTION TO HYPERTROPHY

When it comes to building muscle, there are things that matter and things that don't. There are things I will discover after this manual is written that I implement into my training. There are things we don't have the answers to yet. 50 years from now some of the things we currently believe will change. That is what makes science based lifting both entertaining and frustrating. People will often speak in black and white on topics that are gray. Likewise, there are others who will make topics gray that are actually black and white.

My goal behind this manual is to help you progress. No matter your current level. This is not a scientific textbook. I am not going to be referencing studies every other sentence. Many of these concepts are my own amalgamations of the body of literature and this is not a graded essay. It is simply meant to be a guide to help you build muscle more quickly and efficiently than you are currently. There are some things you may disagree with, and

that is completely fine. If you've spent a good amount of time in the science based sphere, some, or even all of these concepts may be known to you.

Often, people spend more time quibbling over the minute details rather than basic/simple concepts that are far more substantive to the development of your physique. That is why this manual will primarily deal with the things that will make the MOST difference. I am not saying the minor details don't matter, but I am saying that focusing all your time and energy into them OVER the more basic concepts will hinder your long term growth. If you are new to lifting and there are terms you don't understand, most should be easily searchable, but I will do my best to keep the language manageable.

We will start with concepts and build our way to practical application as we go.

Progressive Overload

Without this, you have nothing. Every gym has men in their 30s/40s lifting very light weight despite working out since they were in highschool. And their physiques reflect this. Unfortunately it has become popular to say that strength and size are completely different. This is NOT true. Do they overlap perfectly? Of course not. But there's a very good reason you are more surprised when you see someone 150 lb benching 405 than someone 250 lb benching the same weight. Deep down we know there is huge overlap, even if we don't care to admit it.

In fact, there is such a large overlap that powerlifting MUST have weight classes (and a lot of them) in order for the sport to be available to a large number of people. However, people often throw out the huge overlap because of a few poorly placed anecdotes. "Well I saw so and so with huge biceps curling 25lb dumbbells and he has HUGE arms". Sound familiar? Of course it does. But the statement lacks context and nuance and leads to the wrong conclusions.

Not everyone agrees on what I'm about to say next, but for the sake of keeping things simple, I am going to use the definition for progressive overload that I believe makes the most practical sense.

PROGRESSIVE OVERLOAD:

An increase in the amount of weight, reps, or both on an exercise completed by the same individual with STANDARDIZED technique

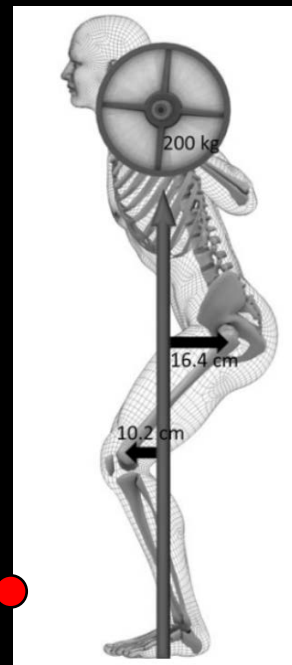
If you curl 30lb dumbbells for 5 reps to failure in a session with a 1 second concentric (positive portion of the movement) and a 2 second eccentric (negative portion), you would need to increase to 6 reps or more the next session or complete 5 reps the next session with more weight and the same tempo to be considered progressive overload. This does NOT mean there has been no improvement if you do the same number of reps and weight. It simply means in this context you would not call it progressive overload.

For example, in the previous scenario if you performed the dumbbell curls the following session for 5 reps with the same weight and tempo, but there were noticeable differences in the amount of effort needed to perform the task, that can also be a sign of improvement. But to keep things as simple as possible, we are going to define progressive overload as simply the increased reps, weight, or both from here on out.

This concept pretty much explains itself. It merely refers to the number of additional reps you could have performed if you were to take a set to failure. You can think of ORIR as failure. I know that's a bit contentious to some, but it will keep things simpler for us, especially when programming. If you did 9 reps but could have attempted a 10th that would be 1RIR. 8 Reps would be 2RIR. So on and so forth. The part where people get the most confused is if they perform 10 reps but "know" they wouldn't be able to get another. Is this 0RIR or 1RIR. Logically it would make sense to think of this as 0RIR because there were "no" reps in reserve. However, the argument could be made that you never "truly" know that you couldn't get another until you try and fail.

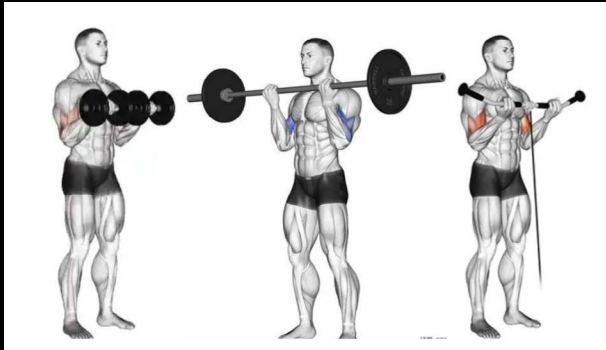
ROM (Range of Motion)

Telling the subject that they must fully straighten the legs after this point to achieve Full ROM is arbitrary and likely unnecessary, as they could directly reverse their momentum into another rep. That is why you may hear me refer to some movements as “Arbitrary ROM”. Fully straightening the legs above a point



where the acting tissues are provided any meaningful stimulus would be an example of Arbitrary ROM. In general Full ROM can be used when in doubt, but it will not be necessary for all muscle groups at all times to achieve an adequate stimulus.

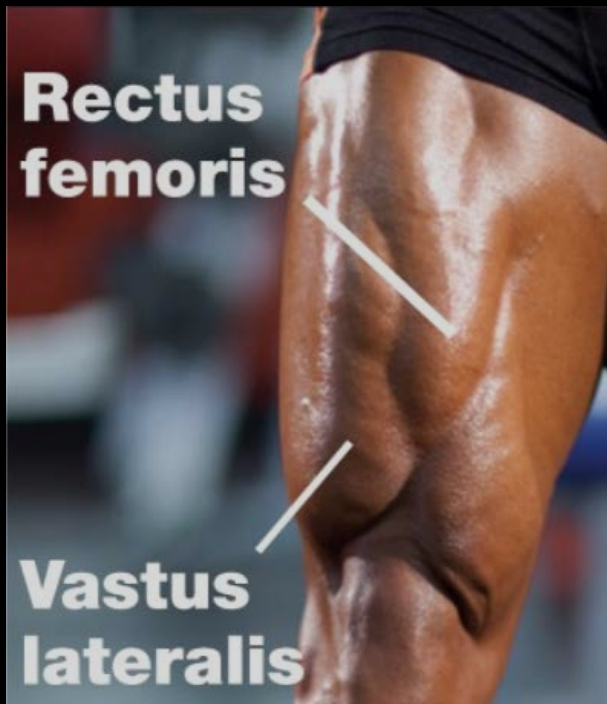
Redundancy



Everyone has seen the person in their gym that walks in, grabs a barbell and does 3 sets of curls before moving to the dumbbell station, doing 3 sets of curls, and then finishing off with straight bar cable curls for 3 sets. This would be a good example of exercise redundancy. You are challenging the same muscle group through a very similar movement pattern.



This does NOT mean that you can never work the same muscle group in different ways during a session. If you decide to do a pressing motion for the quads and a leg extension in the same session we would not consider this complete redundancy. Or at the very least I would not. While they both emphasize quadriceps, they have more tension in different portions of the movement and even work slightly different muscles. The Rectus Femoris, which is one of the heads of the quadriceps, is incapable of being worked meaningfully during pressing motions but will work very well in extensions.



| Intensity

In a hypertrophic context intensity refers to proximity to failure. The closer to failure, the higher the intensity of the set. 5RIR or more would be considered low intensity by almost everyone in the science based community. People may quibble over what they would consider to be “high intensity”, but at the very least we can agree that getting closer to failure is progressively higher intensity with 0RIR most certainly being “high intensity”.

| Volume

Volume in a hypertrophic context is always referring to the number of working sets done. This is not to be confused with volume load which is the total amount of weight moved in a workout. So if I say that my volume for the biceps is 4 sets this means 4 sets WITH high intensity. Warm up sets are not counted towards volume. A working set is defined as being within a predetermined proximity to failure. If the RIR is very high, it should NOT be counted as a working set. That would count towards warm ups.

Stability

This is a highly contentious topic (but shouldn't be) within the bodybuilding community. Generally speaking, stability is something that you should seek out within an exercise if hypertrophy is the goal. And deep down every bodybuilder knows this. If you ask someone the best way to grow their back and

they give you this exercise...



You would automatically know their advice is garbage. So the TRUE point of contention is not whether stability matters. It's how much it matters. We would all likely agree that

this...

is worse for quadriceps growth

than this...



But if you were to claim that the regular barbell backsquat is worse for quad growth than this...

you will quickly find yourself in an argument with many people. The hack squat is **OBJECTIVELY** more stable, but is it **BETTER** for hypertrophy? Provided the user understands proper foot placement and movement patterns, I would argue yes. You're free to disagree, but at the very least I hope this causes some level of thinking.

In MOST cases, additional stability is a good thing for muscle growth. It allows us to more intentionally focus on the muscle group we want to grow and less on coordinating the movement. You see, people often separate brain and muscle, but the two are intrinsically linked. If the brain is spending too much of its attention on stabilizing the body, it will actively take away from the stimulus we are working for. This does NOT mean that you can't grow a muscle without perfect stability. It simply means that, generally speaking, it makes more sense to pick the more stable exercise IF all else is equal (path, resistance profile, comfort).

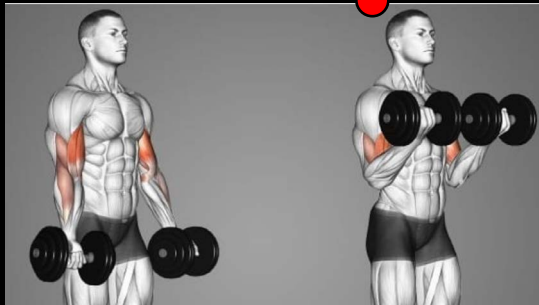


This is why you will see people within the science based community opt for quality machines over free weights in many circumstances. This is not to say that the use of free weights will not lead to growth. In fact it would be moronic to say that it won't. But if the movement pattern and construction of a machine is biomechanically solid, it is difficult to make the argument to choose the free weight option for maximum efficiency. Having said that, at times the construction of the machine can be imperfect leading to the free weight option being a higher quality choice despite increased stability demands. How this looks in your home gym will be individualized.

Isolation vs Compound Exercises

Similar to the stability conversation, in most cases isolation is the better choice for maximum growth of specific tissues. And again, most people agree on this until it directly interferes with something they have traditionally done.

For example, the vast majority of bodybuilders agree **that this...**



would lead to more biceps growth **than this**



But there will remain heavy debate on other compound vs isolation exercises for muscle growth until the end of time. In fact, even the example above has been challenged by many people like Mentzer and followers, though that idea has lost popularity over time. Perhaps by the time you read this manual, it will have picked up again. Only time will tell. Long story short, compound exercises and isolation exercises can BOTH be used with great effectiveness, but typically to see max growth within a muscle group, it would be preferable to attempt to isolate at some point or another.

PS. I'm sure there will be a "well ackshually" from someone reading this and yes, it is impossible to fully isolate any muscle. More like "severe biasing". We good?

Frequency

If I could choose one topic that has the most buzz surrounding it at the time of creating this, it would be frequency. It currently is HIGHLY contentious and I will attempt to be as impartial as one can while discussing it. To understand this discussion, I probably should give you a couple studies that influenced people's thinking on this topic. In 2016 this paper was released by Brad Schoenfeld et al.

I will not go into every detail of the study,

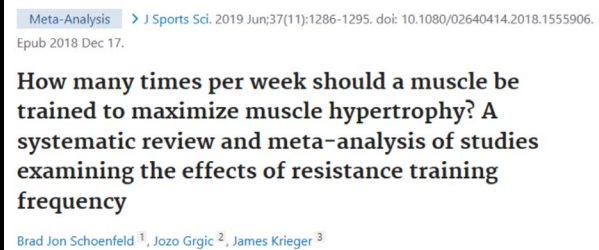


but the practical takeaway for many as stated in the conclusion was this:

CONCLUSIONS:

When comparing studies that investigated training muscle groups between 1 to 3 days per week on volume-equated basis, the current body of evidence indicates that frequencies of training twice a week promote superior hypertrophic outcomes to once a week. It can therefore be inferred that the major muscle groups should be trained at least twice a week to maximize muscle growth; whether training a muscle group three times per week is superior to a twice-per-week protocol remains to be determined.

Seemed pretty straightforward and for at least a few years the majority of people agreed that training a muscle group with a frequency of more than one time per week yielded superior results. However, just a couple years later, a paper by the SAME author came out.



Contradicting the previous review, this paper stated in the abstract:

“Results showed no significant difference between higher and lower frequency on a volume-equated basis.”

And gave the practical takeaway:

“Thus, for a given training volume, individuals can choose a weekly frequency per muscle groups based on personal preference.”

In other words, the update was that overall volume per week was what mattered and NOT whether a muscle group was being trained multiple times within the training week. This threw people for a loop. Bros rejoiced saying “See! We told you bro splits are the best”. Science based lifters rejoiced saying “See! Volume is super important” and the world went on.

Weekly volume became the main metric that science based lifters used for muscles. As long as an overall number was hit within the week, people were satisfied. Alas, it was only the beginning of the war. People began to question these findings. Certain mechanisms weren't adding up. Here are some of the questions people began to have:

If weekly volume is all that matters, why do we have a study showing superior results for training a muscle 3x/week with 1 set compared to 1x/week with three sets?

If we know that growth from an individual workout does not last seven days, what is happening for the remainder of the week?

Why do strength gains seem to level off much quicker than hypertrophy gains in high volume studies? If that much more contractile tissue was added, should strength gains not continue to rise alongside the additional growth?

These questions/mechanistic issues have led many people to believe (including myself) that the takeaway from this review must be wrong. That frequency matters and weekly volume is not the highest quality metric to make the foundation of hypertrophy programming. And that leads us to our next topic.

Sequencing

Imagine you're a track athlete and I told you to run 100m as fast as you possibly could. Let's say you achieve a time of 13.5 seconds. I then give you 5 min to recover. Your heart rate goes down, your muscles aren't sore and overall you feel good. I then ask you to do it again. Do you think your time would improve? Assuming you did things right the first time around, you would not expect to run faster the second time around. In fact you would expect to be slightly slower. Despite feeling good, fatiguing mechanisms are still present, and this reduces output.

Even if you did happen to run as fast the second time due to an unforeseen variable, do you think you could do it again the 3rd time? How about the 4th? Slowly but surely you would be forced to admit that your body just isn't capable of putting out a max effort repeatedly, even with enough rest time for the heart to slow and the muscles to "feel" relaxed. We fundamentally understand this when it comes to sport, but fail to apply it at the gym.

If you had 8 total exercises to perform in the day and biceps were the muscle group you wanted to grow the most, it would be foolish to make biceps the 8th exercise in the day. Fatiguing mechanisms lead to decreased output and ultimately less growth. Whenever you are deciding how to program your workouts, sequence based on the goals for your physique. Of course genetics will mean that some muscles grow easier than others. But if your strong points are being sequenced first in your workouts, they will even more significantly outpace your weak points than they already are.

Fatigue is why this next topic matters so much.

- 1 Without training, EVERYTHING is a weak point. You haven't seen what your genetics hold in store for you. Some muscle groups will grow very fast initially even if they SEEM like weak points right now.
- 2 Being new to the gym, you likely don't have an eye for what weak and strong points even look like.
- 3 Focusing on smaller muscle groups like the posterior deltoid when you haven't grown the huge amounts of tissue you can through focusing on larger muscle groups (quads, hamstrings, glutes, lats, traps, etc) makes no sense. Focus on overall size first with the muscle groups that have a larger potential for overall growth.

Alright, maybe I wasn't serious about the last two. But many people will split their training regimes up into tight little boxes that encompass certain timeframes. If you are a little OCD and feel the desire to do this, all the power to you. I personally do not. When I am determining how I am

It has become popular for people to argue that routine needs to be changed for a variety of reasons, but if you are progressing in your exercises in a manner that aligns with your goals, do not feel like you need to change just because you've been doing the same programming for a long time. Yes, this could even be for years at a time. Most people will want a shift in their training before a few years pass, and that is totally fine. But theoretically, if your goals did not change and your programming was solid, you could run it for the rest of your life.

HOWEVER, if you find yourself thinking your goals change every two weeks so you are constantly changing your program, this is a mistake. Allow your body to settle into a general routine, and at the very least hold those routines for 2+ months at a time. It's not that you COULDN'T make progress switching your program up every other week, but it is wiser from a fatigue, repeated bout, and simple logging standpoint to keep things close to the same for a solid period of time (AT LEAST a few months).

A straight set is a working set that one performs with a standardized technique and weight from the beginning to the end of the set. These are sets taken to or close to failure. Failure occurs when a rep cannot be completed with the **SAME** technique as the first rep. If the ROM is **SIGNIFICANTLY** less or the person needs to change their technique **SIGNIFICANTLY** in order to complete a rep (cheat), they have reached failure. The reason I am putting the word significantly in bold is because many people will try to be so perfect that they will overthink very slight changes in range of motion or technique that definitely would not be considered failure and bail a set.

If you are ever unsure, film yourself doing a set and watch it back. You can typically see if form broke down too significantly or range of motion changed too much. This will give you solid feedback for next time.

Dropsets, Supersets, MyoReps, Mechanical Dropsets...you've probably heard many, if not all of these terms thrown around at some point or another (yes, even by yours truly). The more we learn about fatigue and stimulus, the more we are seeing a move away from regularly incorporating intensity techniques by many within the scientific lifting sphere. The reasoning is fairly simple. Nearly all (if not all) intensity techniques do not seem to add any meaningful stimulus that could not simply be accomplished through adding additional straight sets.

TNF NOTE THAT I AM COMPLETELY WILLING TO BE WRONG ON:

Those that are currently undershooting their volume by a significant amount would actually see more progress by turning some of their straight sets into sets involving intensifiers. Reason being that they have not reached the max stimulus for some muscle groups and would therefore benefit from additional stimulus EVEN IF it is coming through an intensity technique.

Now you may ask why I mention this as those individuals could add more straight sets in rather than doing intensifiers. That is a fair question and I'm glad you asked, astute reader. Some people are limited on time or simply prefer to throw intensity techniques into their training. I want it to be well understood that those who choose to do so can still make fantastic progress.

Straight set: a set that is taken to or close to failure with no additional qualifiers.

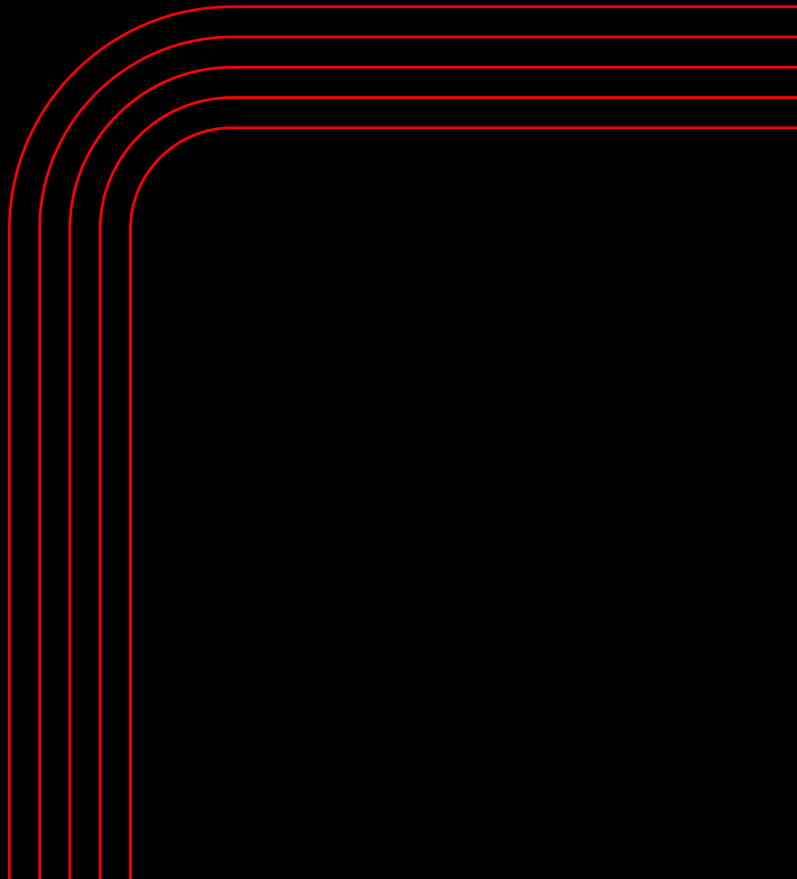
Dropset: A set that is taken to or close to failure, subject then reduces load and continues to or close to failure again. This process can be repeated however many times the subject desires.

Superset: Where two exercises are performed back to back with little to no rest in between.

MyoReps: A set where the subject chooses not to reach task failure, rests a brief time, and continues the set with the SAME amount of weight. This continues for a predetermined number of times (typically getting closer to or to failure by the end).

Mechanical Dropset: an exercise that remains the same but is mechanically loaded in different positions throughout the set.

With a more thorough understanding of Motor Unit Recruitment, it appears that it would be difficult to justify doing many intensity techniques if the goal was MAXIMUM growth. However, as a timesaver, or for pure enjoyment, we do have plenty of data to suggest intensity techniques can still be used efficiently.



HOW TO ACTUALLY PROGRAM FREQUENCY

Now that we've gone over a bunch of topics, the real question is how do we APPLY what we know. We could have skipped straight to this step, but many people would rather understand where these concepts are derived from. First thing we need to decide is what general frequency you're going to do for a muscle group. The overall body of literature seems to suggest that working a muscle group more than once a week is preferable for growth. There is heavy debate within the science based community on whether 2x or 3x is better. Some even argue for hitting a muscle group every other day (3.5x frequency).

In the end, your safest bet is going to be a training frequency that you enjoy and gives adequate stimulus and rest.

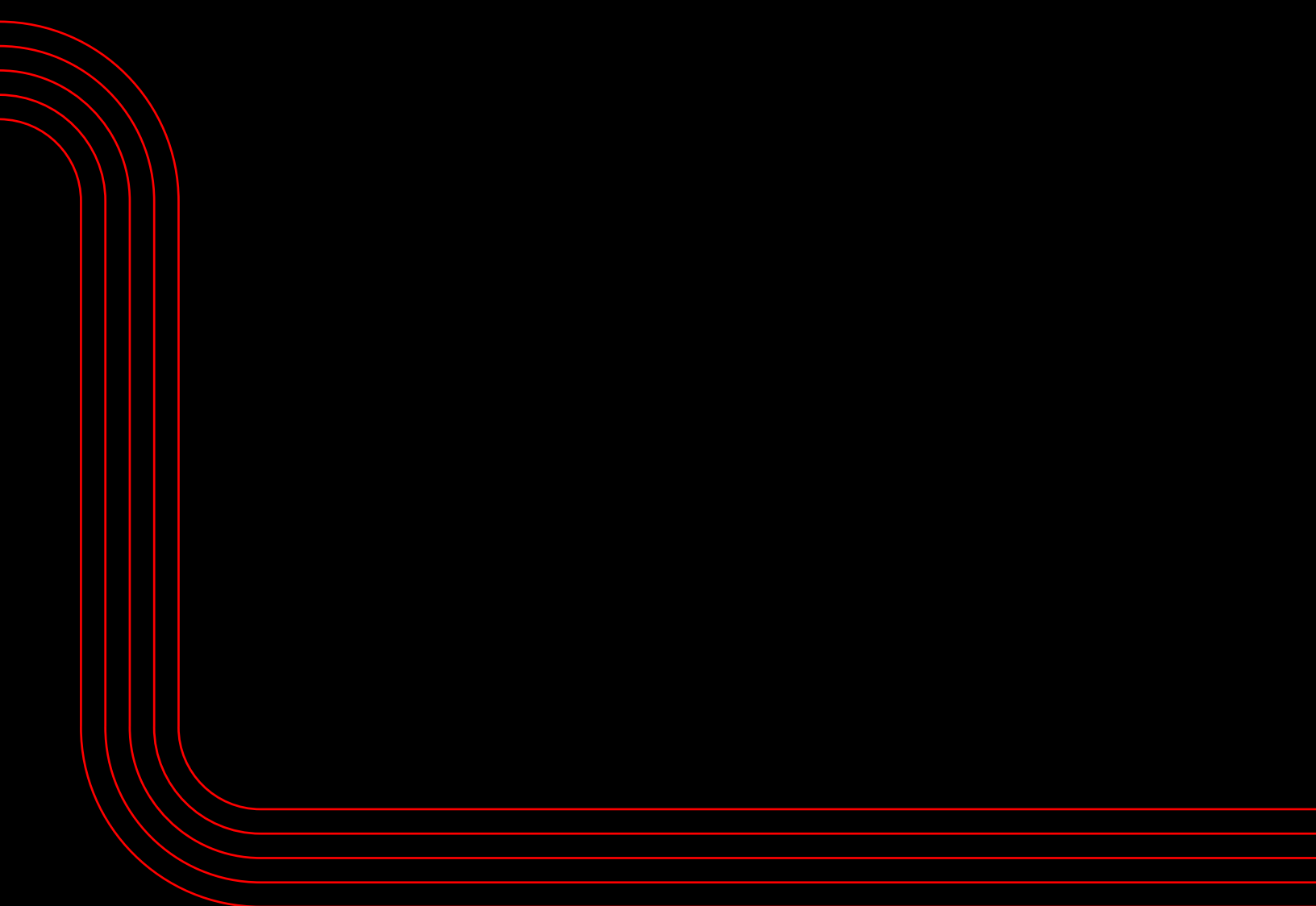
If this answer does not satisfy you, I apologize but based on the current available mechanistic AND outcome data I will not give you a firmer stance. Both can be done with VERY high efficiency. I would argue

that even 1x/week training has shown solid efficacy given adequate volume (sorry to everyone who wants me to say that you will grow zero muscle and implode with 1x/week frequency)

However if we look at things objectively (especially with lower volume training) it becomes very tough to argue 1x/week being able to compete with 2 or 3 times per week frequency. Which brings me to my next important point. How many sets I suggest you program for a muscle group will change based on your overall frequency. If you are only training a muscle group 1x/week, I would suggest keeping your volume quite high. Simply put, you will both have a long time to recover and possibly a long time for atrophy if you go 7 full days without a significant stimulus. So while I don't generally advise training muscle groups 1x/week, if you do, I would prefer a rather high volume OR intensity techniques being added to some sets if you are planning on keeping set volume relatively low. I have trained in this format

at various points in my career with great effectiveness. I preferred not to do endless sets, so I instead would do 5-6ish sets for the muscle group with added dropsets, mechanical dropsets, or partials. To say that I didn't grow while training in this format would be a lie, even if I don't currently consider it to be the most ideal way to train. And this is something that a large number of "science based" creators totally miss the mark on. If you have many variables right, it likely won't make an insane difference to have a couple that aren't absolute perfection.

With that sidenote out of the way, training with more than 1x/week frequency (if smartly managed) will likely lead to better results for the majority of lifters. This does NOT have to be exactly 2 or 3x frequency. As we alluded to before, some do 3.5x frequency or even higher. I myself have done decimal points between 2 and 3 or 1 and 2 at various times. There is nothing inherently special about 7 days and your split does NOT need to be based around a calendar week. It is the format that most people adopt because humans like organization, but it is by no means a necessity.



HOW TO ACTUALLY PROGRAM PROGRAM VOLUME

As we hinted at before, volume is fundamentally based on frequency. Imagine I told you to do 30 sets of bicep curls and then told you you would need to do that every other day. This would lead to a frequency of 3.5x/week and overall weekly volume of 105 sets. Despite what some would have you believe, I cannot in any world imagine this to be effective training. And while I would still consider you crazy if you programmed biceps 1x/week with 30 sets, it would make far more sense than the prior option.

In other words, your volume is going to be HIGHLY dependent on your frequency with the two being inversely proportional (As volume per day goes up, frequency must come down). My guideline in the majority of cases (assuming sets taken to 0-1RIR) would be as follows:

- 3x/week frequency or more: 1-3 direct sets/day
- 2x/week frequency: 2-6 direct sets/day
- 1x/week frequency: 6+ (can obviously be quite high)

This may seem quite low (and compared to a lot of bodybuilding history it is) but I would contend that there are many reasons this can be highly effective. One reason is that many people simply don't train as intensely as they believe and therefore miss out on stimulus on a set per set basis. And the other is that I'm referring to

volume towards every muscle group listed other than the calves. 3 movements, 11 muscle groups. This would be the minimalist way of viewing programming. 1-2 sets in a day for each movement and you could be doing as little as 15-20 total working sets in a week.

As with most things, I personally believe that the true solution for the majority of people is going to lie somewhere in the middle. If the literature has taught us anything about maximizing muscle growth, much to the chagrin of absolutists, it has taught us that there are a number of very efficient ways to get to the same goal (huge muscles). My goal is to help you in the simplest ways I can to achieve a bodybuilder worthy physique. And don't worry if you have zero aspirations to become a competitor. You know what you do if you don't want to "get too big". Stop progressively overloading once you're happy with your size. But until you achieve that size (spoiler-it's probably going to take way more work than you think) you better be training and eating like you want to be the next Mr O.

next Mr O.

Here we are looking at a MINIMUM of 12 muscle groups and, if isolated/severely biased, 12 exercises. Automatically that would mean 36 total sets in a week for the 3x frequency MINIMUM and going over 100 total weekly sets if 3 direct sets were done each day. Add to this that many people who are currently attempting to isolate every possible exercise are also doing a majority of their exercises unilaterally AND doing multiple movements per muscle group for even more severe biasing. This is a maximalist way of viewing programming and can easily lead to hundreds of working sets.

That being said, part of the reason compound movements are so heavily relied upon is due to their ability to reduce overall set volume. Imagine if someone does a squat pattern and counts that towards glute, adductor and quadriceps volume. Throw in a chest press and a non chest supported row and you've now (in many people's minds) put at least some

PROGRAMMING AS A BEGINNER

If you've purchased this manual there's a high likelihood that you're not a complete beginner in the gym. But even if you aren't, we should still talk through the principles of being a beginner lifter in order to help you understand how to view programming as you get further along. Here is the most fundamental concept to understand. When you are completely new to the gym, muscle growth is EASY. Very basic guidelines can help you to pack on a lot of muscle. This is often why you'll hear advanced lifters say beginners need to "build their base" with very particular exercises like the squat, bench, and deadlift.

It is a very dumbed down way of saying that they'll gain plenty of muscle in the beginning if they just focus on big movements where the goal is to move a lot of weight with a large number of muscle groups. The craziest thing is that they're not far off the mark. While we're not going to get into Motor Unit Deficits (you can look this up on your own time if interested), the basic principle is new lifters just aren't that good at recruiting all of their muscle. So spending the majority of time in hyper specialization (unilateral single joint exercises) in my opinion doesn't make much sense.

I would much rather see a new lifter program 15-20 sets of compound movements in a training day and perhaps a few isolation sets (still bilateral) than some of the maximalist routines that are currently being performed. Here's a sample upper day that could make sense for someone new to the gym. All sets within 0-1RIR.

- | | |
|------------------------|--------|
| 1. Flat Chest Press | 4x6-10 |
| 2. Chest Supported Row | 4x6-10 |
| 3. Shoulder Press | 4x6-10 |
| 4. Lat Pulldown | 4x6-10 |
| 5. Lateral Raise | 2x8-12 |
| 6. Bicep Curl | 2x8-12 |
| 7. Tricep Extension | 2x8-12 |

And I'm sure many of you reading this are surprised at seeing a recommendation of 4 sets for exercises. Or the possibility of reps going above 10. But if you are new to the gym and you ran this exact upper day, you would gain A LOT of muscle in your first year of lifting. Does this mean that this structure is perfect for everyone? Of course not. But to deny that this would work efficiently for most beginners would be ignoring everything we see within the literature and anecdotal practices.

This is arguably tougher to manage than an advanced athlete. There's a couple of reasons for this.

- 1 Anybody who's been in the gym for more than a few weeks assumes they're intermediate (they're not)
- 2 The true intermediates are often thinking they are advanced (they're not)
- 3 Intermediates assume they need highly advanced and specific training (they don't)

So how do you know if you're intermediate? IF you have been in the gym for a couple of years and IF you've been training in a relatively smart manner, you would likely be getting into the intermediate phase. This should be clearly noticeable in your physique. If you tell me you're an intermediate lifter and you've only built 10lb of muscle since you stepped into the gym (barring extreme circumstances) I'm going to look you dead in the eyes and tell you you're still a beginner and should view training through that lens.

Now that we've made everyone question whether they are truly intermediate, major differences in programming are relatively minor on paper. The volume you COULD handle as a beginner is harder to handle as an intermediate. People often take that as me saying you should automatically reduce your volume as an intermediate lifter. Not so. Perhaps you could have done more volume as a beginner lifter but simply didn't so there is no volume change going into your intermediate years. However, if you were maxing out volume in your first couple years you will almost certainly have to tone it back.

This often confuses people as they typically believe that in order to keep growing the volume has to go up not down. But as you become stronger and more capable of recruiting muscle to its fullest capacity, you will also be more capable of overdoing things from a damage and fatigue standpoint. When sprinters are young they can run 100m as fast as they want and feel good to repeat that endeavor the next day, and the day after that. Professional sprinters know they cannot behave in the same manner. They would rip themselves apart. While the risks from slightly overdoing volume are not as great for lifters, they still exist.

Now onto how I think of programming for an intermediate (you're perfectly free to disagree). Intermediates need to be a little more specific with their goals to keep progress coming quickly. Progress will inevitably be faster the first couple years even with basic programming, but intermediates might

PROGRAMMING AS AN INTERMEDIATE

want to branch into biasing a little more heavily. This is when you start to think about weakpoints, managing fatigue a little more closely, heavier emphasis on isolation movements...etc.

EVEN AS AN INTERMEDIATE, I think people attempting to isolate every single joint action are going overboard. A solid blend of heavier compounds and isolation movements make sense for an athlete in this phase. If you noticed with the beginner athlete we did not attempt to isolate any of the trunk muscles within the upper day. Shoulders, Biceps and Triceps were the only muscle groups that were given isolation. Now as an intermediate I think we can explore the idea of isolating trunk muscles as well. This does NOT mean that you must isolate everything that can be isolated. However, if you notice that the traps are behind other muscle groups, I personally believe adding in shrug variations would make sense. So let's look at what upper programming MAY look like for said individual.

- | | |
|---------------------------------|--------|
| 1. Incline Chest Press | 2x5-8 |
| 2. Lat Pulldown/Weighted Pullup | 2x6-10 |
| 3. Pec Fly | 2x5-8 |
| 4. Chest Supported Row | 2x5-8 |
| 5. Lateral Raise | 3x5-8 |
| 6. Preacher Curl | 2x5-8 |
| 7. Tricep Extension Underhead | 2x5-8 |
| 8. Tricep Extension overhead | 1x5-8 |

"But Joel this is over X sets in a day and that means that the last sets won't be productive!" No, I don't ascribe to that thought process. I don't believe there has been a valid study demonstrating this and if you happen to be a person who believes they personally need to stay under a certain amount of sets, that is fine. Just don't mistake your personal situation for the standard. And if you do find yourself in that situation, the solution is relatively simple. Reduce volume for the exercises.

This will likely involve additional specialization if you believe certain muscle groups are behind. Can you apply some of these principles as an intermediate athlete? Of course. But the law of diminishing returns suggests that specialization will always matter more the further you get into your training career. HOWEVER, even now there are many people specializing beyond what I believe to be necessary. And at times they will watch how I lift and scoff. That's completely fine. I simply don't believe every muscle group needs to be specialized to such an extent that you end up with 50-80 different exercises in your program (no that's not a joke, some people train this way)

The easiest way to understand how training changes as you move into this phase of your career is to simply think about a few variables and how it will change your programming. Again these principles will somewhat apply to intermediate lifters as well, just not to the same extent.

Volume:

Advanced lifters will have a harder time handling intense workouts with very high volume than people who are earlier in their training career

Intensity:

While advanced lifters still need to train with intensity, they must also be very aware of how training with too high of intensity too often (failure, partials, intensity techniques, etc) can limit their ability to progress consistently

Isolation/Severe Biasing:

This variable becomes more important to continue to grow as an athlete gets more advanced. Can they still grow with large compound lifts? Sure. But the further you push into your training career the harder it becomes.

Stability:

This variable also becomes increasingly important as you get further in your career. No this does not mean you can't grow with freeweights. It simply means that leaning towards more and more stable movements makes sense.

PROGRAMMING AS AN ADVANCED ATHLETE

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You get the picture...

From MY perspective, it is unreasonable to ask people to try to train all these components in heavily biased/isolated fashions. The rise of the maximalists has been interesting though and I'm curious to see if they'll continue in this fashion and what types of physiques they'll build with it. Maybe they'll pioneer in a new age. I simply have my doubts.

Using the original list that I made, advanced athletes should look at what they want to prioritize and write out a list from strongest to weakest points OR most important to least important desire for growth. For example, that list for myself might look something like this:

- Lats
- Chest
- Quads
- Hamstrings
- Shoulders
- Traps
- Biceps (Elbow Flexors)
- Triceps
- Calves
- Adductors
- Erectors
- Glutes
- Abs
- Forearms

All that this means is that in my current thinking for programming, lats are the number 1 priority and forearms are the least of my concerns. Your program needs to reflect your priorities as an advanced athlete. Here is my preferred thought process (you can develop your own if you want) for how I prioritize muscle groups within my programming.

Exercise selection (stable/loadable/more severe isolation) → Order (earlier placement in workouts) → Volume (likely going to have higher volume than groups that are not the priority) → Regions (if there is a way to bias different portions of the muscle I will ensure a variety of exercises to achieve that purpose) → Unilateral movements (can possibly lead to increased motor unit recruitment depending on the exercise and muscle group)

Obviously other than Order, you could do all of these things for all muscle groups. Increase volume, unilateral movements, multiple movements for more biasing within the muscle region...but a program that sought to maximize every muscle in this manner would end up ridiculously long and be impossible to recover from. The body can only handle so much.

This is where you have to be smart on when and where to specialize. And this is why listing out your own priorities is so important. To make this more clear, let's imagine a "Back+Bi Day" together. I know it's scary to talk about a Pull day in 2025, but I promise you it can be done in a smart manner. Regardless, the same principles that I apply to this day can be applied to any other day in whatever programming you decide to run with.

First list out the muscle groups you want to train in order. For the breakdown I gave you before it would be Lats → Rear Delts → Traps → Biceps → Erectors

There are a TON of ways you could set up this day, but the most important thing is to take the order into account. If I wrote out the day like this and did bicep curls first, that would be counter to my goals. Similarly, someone who is wanting to prioritize their biceps, but chooses NOT to do bicep curls first because it's a "small muscle group" is also programming counter to their goals. Regardless of what bros have told you in the past, if your priorities dictate it, "small" muscle groups CAN and SHOULD go earlier in your workout.

Now that we've figured out what order we would like to train in, exercise selection is the next step. Here's what a basic structure might look like for this context.

- Lat Pulldowns (Possibly Unilateral)
- Lat Biased Row (Possibly Unilateral)
- Reverse Pec Deck
- Elbow Flared Row
- Preacher Curl
- TNF (Chest Supported Erector Exercise (I don't have a name for it currently))

"Wait, wait, wait...I thought you said this was advanced..." you're right. I am very explicitly stating to every single person that reads this manual that even an advanced athlete could make progress with something like this if they had appropriate intensity and volume for these exercises. Now that we have that out of the way for anybody who likes to be a little more minimalistic in their approach, a CURRENT pull day for me might look something more like this. This is assuming 2x/week frequency.

- | | |
|--|-----------------|
| 1. Unilateral Frontal Pull Motion | 2 sets 0-IRIR |
| 2. Unilateral Sagittal Pull Motion | 2 sets 0-IRIR |
| 3. Elbow Resisted Rear Delt Work | 2 sets 0-IRIR |
| 4. Kelso Shrug | 2 sets 0-IRIR |
| 5. Upright Smith Machine Shrug | 1-2 sets 0-IRIR |
| 6. Elbow Supported Curl (Preacher or Shoulder Extended) | 2 sets 0-1 RIR |
| 7. Humerus Supported Pronated Preacher Curl (descending+ascending profiles) | 1 set ORIR |
| 8. TNF (Chest Supported Erector Exercise (I don't have a name for it currently)) | 2 sets 0-IRIR |

UNDERSTANDING THE THOUGHT PROCESS

Yes, I am completely aware that there are arguments happening as to whether you can even bias the lower and upper lats. Perhaps by the time you read this it will be settled. What we do know for sure is that both of these exercises work the lats well. So I will continue to include both in my programming. They also may be unilateral exercises as lats are my number one priority. Rear delts will be heavily involved in the lat exercises so they will only have 2 sets of direct work (I consider that satisfactory for myself, but you may not). Traps are a complex muscle group and have heavily varied orientation to the fibers. A couple angles makes sense. The elbow flexor muscles are heavily influenced by resistance profiles and supination/pronation so a couple variations for this also makes sense to ensure adequate work for the Brachii, Brachialis, and Brachioradialis. The erectors are not overly important so they come last and have the lowest volume, but still solid exercise selection. Honestly this

exercise likely wouldn't even be necessary in the context of an overall program if I had a heavy hinge on my leg day, but I merely wanted to highlight the erectors being a component of back development.

This workout occupies a middle ground for a lot of the lifting sphere at the moment. Maximalists would say it's not specific enough. Minimalists would say it's too much. Bros would say there's too much isolation...And if you fall into any of these categories, that is completely fine. My goal behind making this manual is not to have you train precisely as I do. But rather to understand how to blend scientific ideas and the gym together in a way that is productive for YOU. This should go without saying, but if you are able to continue progressing these lifts with standardized form, you will continue to make muscular progress even if it isn't perfect (hint; it never will be).

ADDITIONAL CONSIDERATIONS

Not everyone has the same amount of time available or desire to be in the gym. If you convinced me that being in the gym 4hrs/day 7 days a week would lead to the most muscle growth, guess how much time I would spend in the gym per week? 28hrs. So when it comes to intelligent programming you should also consider the cost and reward of your program. I legitimately believe most people could get the vast majority of their gains if they had VERY intelligent programming and were in the gym for 3hrs total per week. I believe they could still get good gains with only 2. I even believe they could get modest gains with 1. So you have to decide which type of person you are. Is the gym a semi interesting hobby, or do you absolutely love being there. Because I would not program the same way for each personality.

We know that 3 min rest between sets is better for growth than 1 minute. But the difference between 2 and 3 is MUCH smaller and possibly nonexistent at times. Is it worth it for you to wait another minute to be sure you're recovered fully? Or does that additional minute added up over the course of 20 sets in a day mean you're spending an additional 1-2hr in the gym per week unnecessarily relative to your goals. Science based lifting is about maximizing goals, not just gains. If you want great gains, you do not need to do nearly as much as if you want ultimate gains.

GENERALIZED PROGRAMS

Now this is what you all came here for. As much as I try to get you guys to think there are some who will simply say “just tell me what to do”. And that’s fine especially as a beginner to intermediate athlete. But at some point I hope you can use some of the principles laid out in this manual to develop your own program. It is truly fun to work through solving the puzzle once you understand the approach.

I will have 3 overarching programs provided for you that I consider to be beginner to intermediate level. Advanced becomes too complex to give as a general program. HOWEVER, I would even venture to say that some advanced athletes could still benefit from some of the simple programs provided here. My only suggestion is to look it over and see how you can further tailor it to your specific needs (special exercise selection, unilateral work, adjusted volume for priorities, etc)

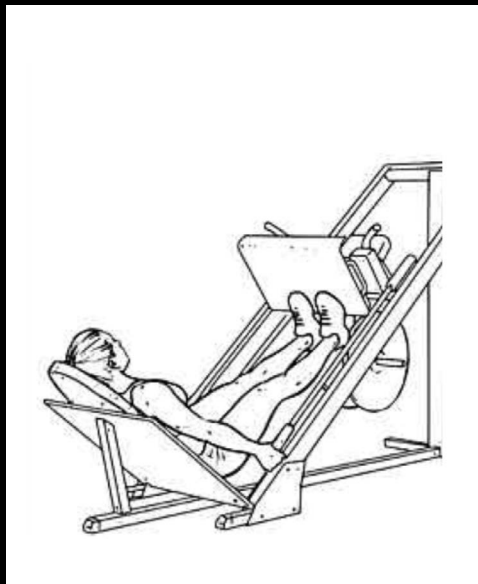
EXERCISE SELECTION

I mulled over how to do this for a very long time. There are too many exercises to go through all, so what I have decided to do instead is go over the major muscle groups and exercises that make sense for them. Some muscles have multiple functions and will need to be trained in multiple ways to grow “maximally”. The only reason that word is in quotations is because you can never truly maximize your growth. But the goal should be to get as close to it as possible. There are also conflicting opinions on certain muscle groups and topics that I do NOT want to take a firm stance on.

As crazy as it sounds this IS the most science based view. Absolutist statements in an area that is anything but absolute are not scientific in the slightest. We should always be open to the possibility of being

wrong. Even though I have written what many in the “science based” community would consider relatively basic takes, some could be wrong. This is a new area and we should view it with trepidation rather than foolhardy arrogance. Now that that is out of the way, let’s go over muscle groups and what I believe science has shown to be effective exercises.

There are also those in the sphere who disagree heavily on some of these topics, so I will attempt to say those I’m familiar with in those circumstances so you can see multiple perspectives. It is best not to be stuck in an echo chamber, but rather to entertain a variety of viewpoints and come to your own conclusions. And if you’d rather not do the thinking, fear not, there will be exercises here that I think everyone can agree are solid.

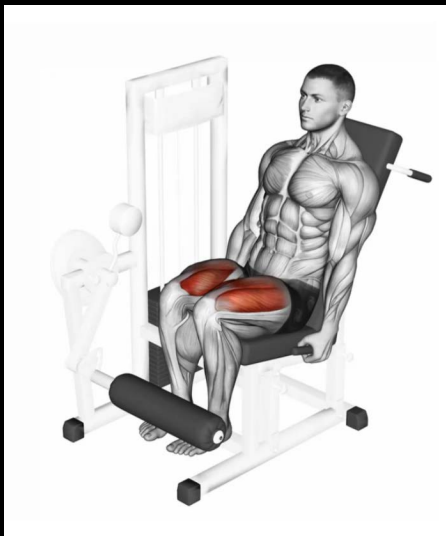
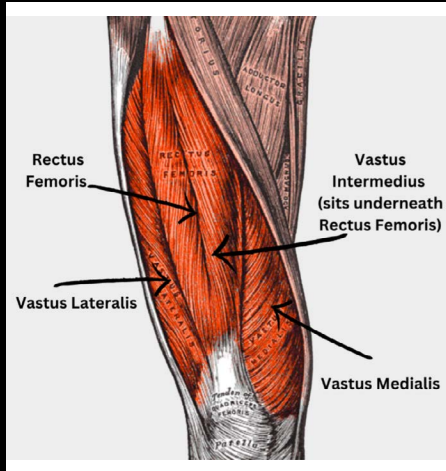


Calves

The calves are relatively simple in my opinion. They are made up of the gastroc and the soleus. While some have argued the benefits of bent knee calf raises, the current literature tends to lean towards superior gastroc growth for straight leg variants. Typically when people are most concerned with aesthetic muscle growth they prefer to grow the gastroc muscles to a greater extent than the soleus. And seeing as the soleus will still grow while training gastroc muscles, many stick to only STRAIGHT leg calf raise variations. However, if you are looking to grow the soleus to its fullest extent, a seated BENT knee calf raise can also make sense. Two images are provided of common calf raise variations. While both are seated, the knee bend is what makes the difference.

There is at least one current study suggesting that lateral head gastroc growth may be greater with toes pointed outwards on a calf raise while medial head growth may be greater with toes pointed in. That being said, you will still get solid growth in both regions with a neutral foot position. If your goal was to emphasize inner or outer growth at some point in your career, it might make sense to incorporate this knowledge.

It is also worth noting that the majority of current literature leans towards favorable growth through completing the BOTTOM half of the range of motion for calf raises (training the calves at longer lengths). This does not mean you CAN'T go all the way up onto the toes. Only that loading at length appears to work very well and it is definitely worth considering doing at least some (if not all) of your calf work in the lower half.



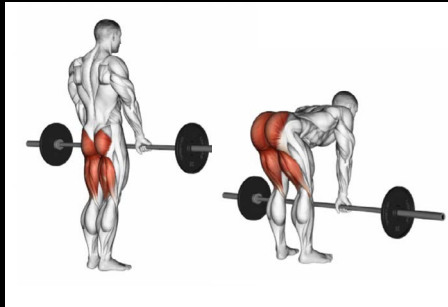
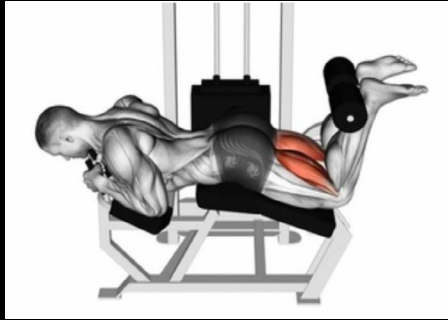
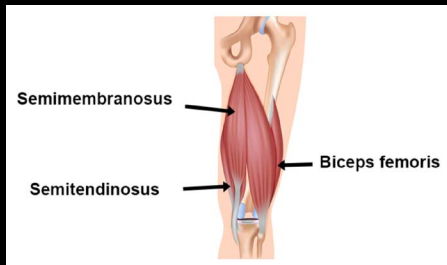
Quadriceps

The quadriceps are relatively simple in theory. The Vastus muscles perform knee extension (straightening your leg) and the Rectus femoris performs knee extension and hip flexion. This means the Rectus femoris is only worked to a significant extent when either the hip or the knee remains stationary while the other flexes (think leg extensions or a leg raise). The vastus muscles will work regardless if the hip is stationary, so long as the knee extends (think squat patterns).

The Vastus muscles also work in exercises like leg extensions which has led to some very bold claims as of late (maybe by the time you read this it will be completely accepted or rejected). Because we are often looking for ways to isolate muscle groups, some have claimed that you may as well drop all pressing movements in favor of solely leg extensions. Because movements like the hack squat, barbell squat, and leg press involve glutes and adductors, they theorize that these movements are not worth performing.

We run into one tiny problem here...the literature. We have had MULTIPLE studies that have found better vastus growth with presses over extensions. The extensions, of course, repeatedly demonstrate better rectus femoris growth. So what would my suggestion be to you? To do what you see fit. I am personally not willing to toss aside results from studies as quickly as many others (they'll even go so far as to make fun of people who take outcome data seriously). There has to be a very solid amount of evidence for me to toss outcomes by the wayside. I am not arrogant enough to believe that we have solved every mystery.

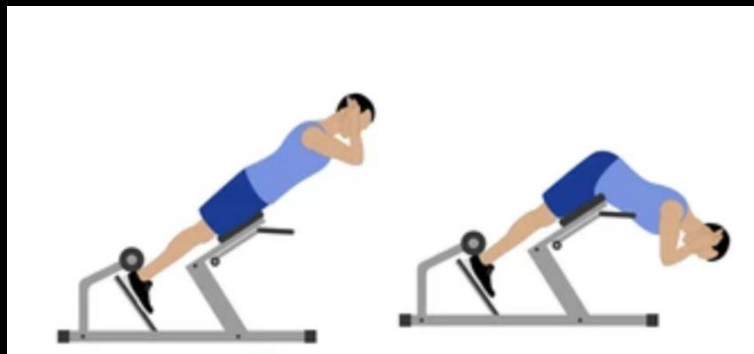
If all the above confused you and you're simply asking what I suggest it would be to have at least one pressing pattern and one extension pattern throughout your career. If we find out that the press was completely unnecessary in the long run, I'm willing to accept that. For now, I will keep them in the rotation.



Hamstrings

One thing is absolutely certain. Curls WORK. Seated hamstring curls appear lead to more overall growth (at least in the short term), but lying hamstring curls also provide a good stimulus especially in particular portions of the hamstrings and perhaps (according to one study) better growth for the sartorius (not a hamstring muscle). So while I will suggest opting for the seated hamstring curl more often or when having to decide between the two, the lying ham curl still has its place in an overall career.

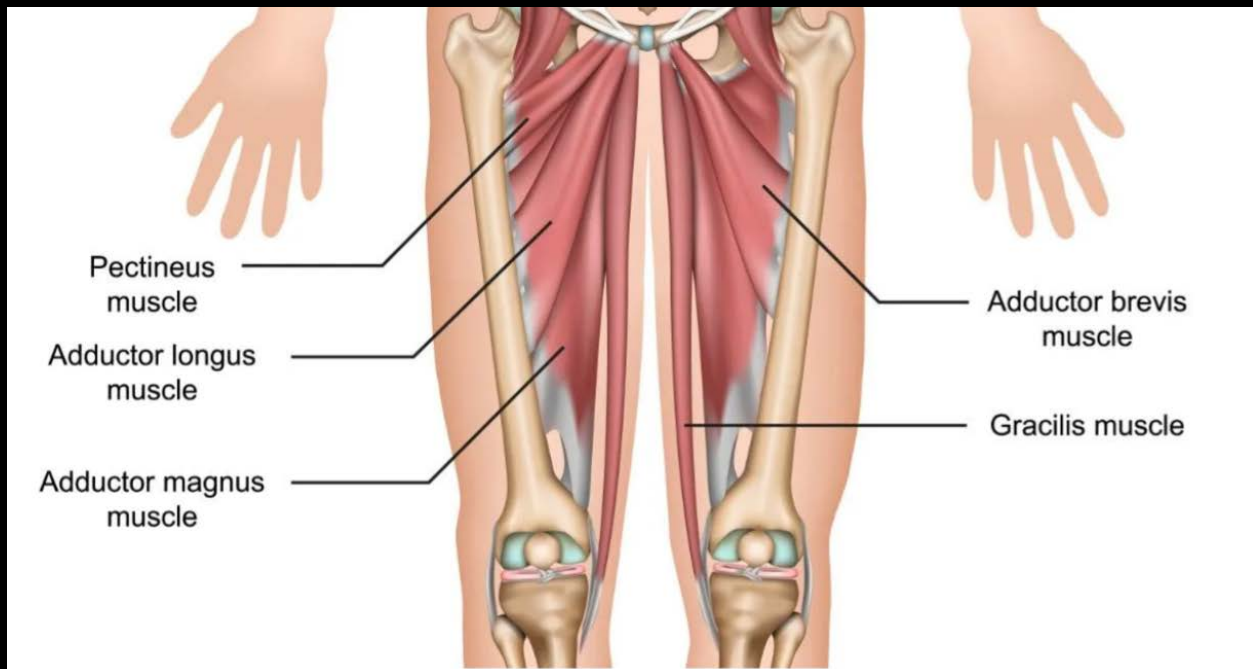
BUT the hamstrings (similar to the rec fem) play a more complicated role. They do not merely flex the knee, but also extend the hip (other than the short head of the biceps femoris). This means that loading with little to no knee bend, but extending the hip will also provide quality stimulus for the hamstrings. Most people have performed deadlift variations for this aspect of the hamstrings, with stiff leg deadlifts being the most popular. However there are other movement patterns like 45 extensions that are also great options.



Due to possibilities in regional hypertrophy differences between hip extension (hinges), and knee flexion (curls) for the hamstrings, I suggest you incorporate both regularly. And you'll see this as a consistent theme. There are often debates when it comes to regional hypertrophy differences from a variety of exercises. My general take in these circumstances is (assuming both movement patterns are good for muscle growth) incorporate multiple movements. You saw this with the quads and now the hamstrings.

Adductors

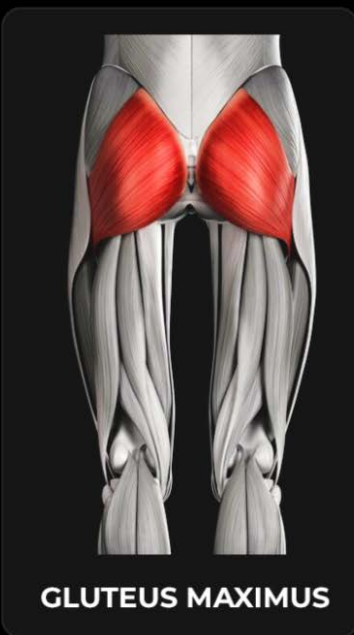
The adductors are heavily involved in many gym movements. They are involved in hip hinges of any kind, and the motion of ADDUCTing your leg. Crazy I know. I'll keep it brief for this muscle group as I don't think it needs belaboring. Your program will almost certainly have a hip hinge of some sort so you won't have to go out of your way for that function. But far and away the most isolated exercise for the adductors is the adductor machine. The one issue you will consistently run into with a variety of adductor machines is loading potential. Many are very light.



There are a few things you can do to fix this problem. The first one is to accept higher reps. While it's not ideal in my opinion to make a regular habit out of 15+ rep sets, you can certainly do it and experience good growth. Another option is adjust yourself in the seat. If the seat has a function to lean back, move as far back as possible and try to make the point of contact with the thigh pad at your knee rather than further up the leg. This will increase the difficulty of the exercise per pound on the machine. And last, you can use an additional pin to load the machine with plates. You should be able to find a pin from another machine at many gyms, but if you prefer you can also purchase something like a Gympin (if cost is an issue I've even seen people bring screwdrivers).

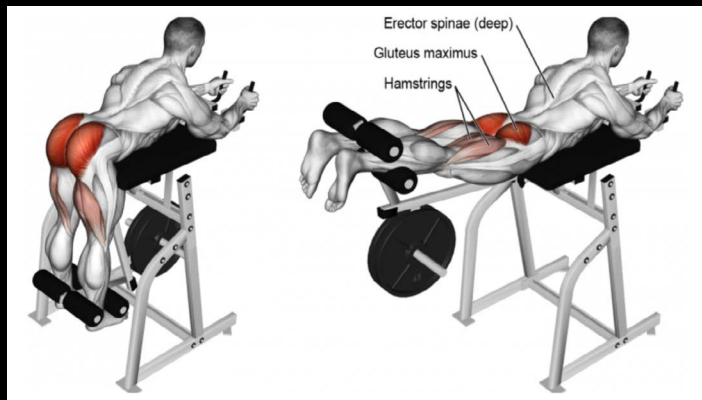
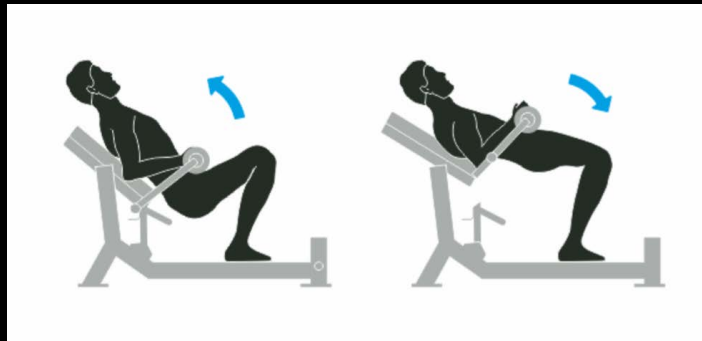
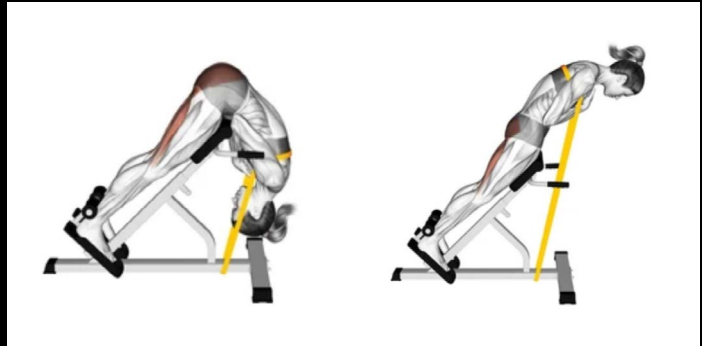


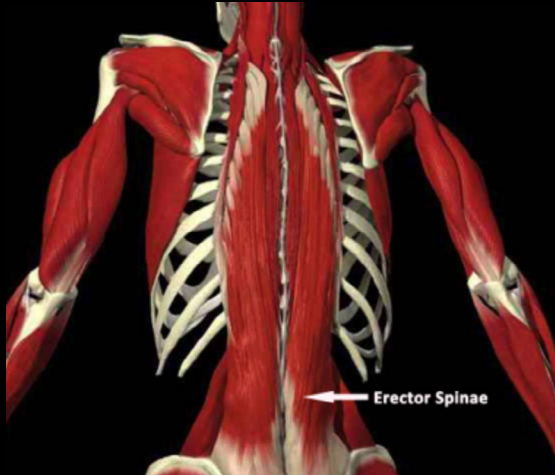
Glutes



Glutes are arguably the most neglected large muscle group in the male fitness world. But this doesn't mean that men don't train them accidentally. Let me explain. The primary function of the gluteus maximus is hip extension. Any hinge pattern (think squat and deadlift patterns) involve hip extension. So, similar to the adductors, most people will be training their glutes to at least some extent within their programming. But if glutes are a focus for you, I suggest getting more specific. Including hip thrusts is the most basic way but there are multiple exercises that can hit the glutes in a shortened position including banded 45s, reverse hyper extensions, and GHD back extensions.

And while I'm not going to spend a lot of time talking through all the intricacies of the gluteus muscles, there are also abduction roles to the glutes which is why you'll see people perform various rotational exercises or use the abductor machine both leaned forward and back. If you want to look into those further to maximally develop your glutes I suggest looking into people like Bret Contreras as this is definitely more of his specialty than mine. But for quality glute development in general I would suggest a hinge pattern, a short position flexed movement, and possibly some abductor machine movements. If those were the only three you did for the rest of your life I believe you would develop some very thick glutes.





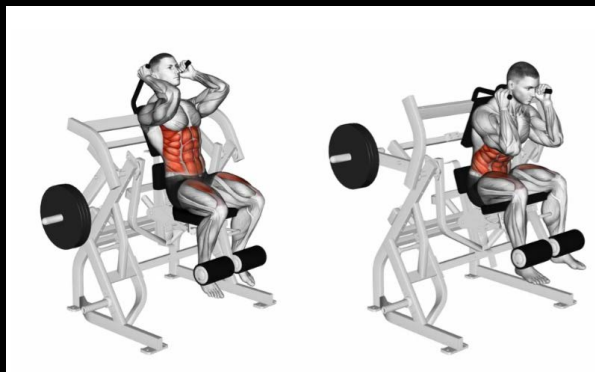
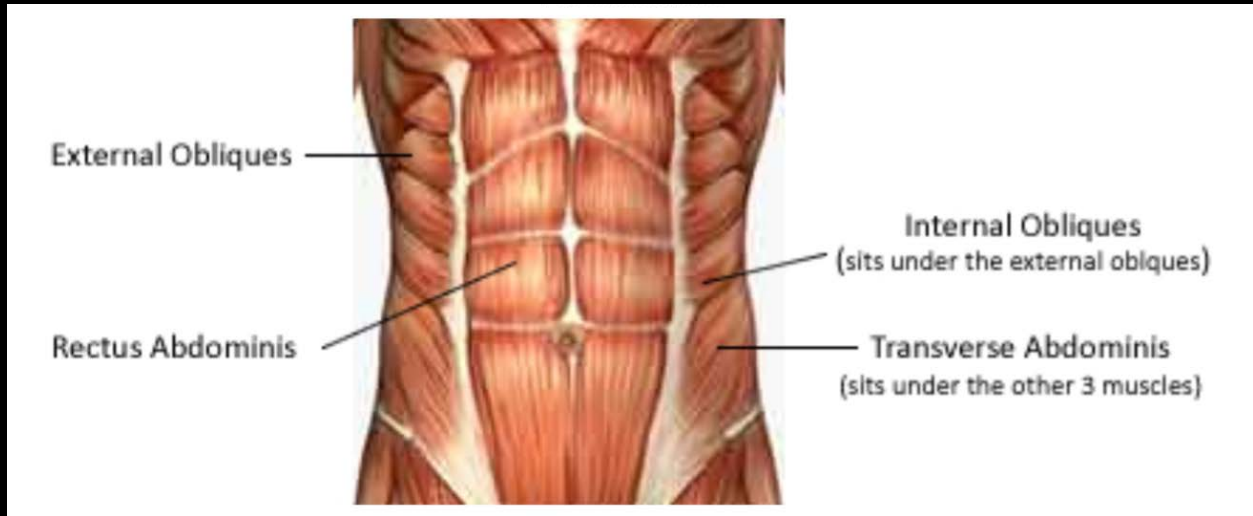
Erectors

One of the most heavily misunderstood muscle groups is the erectors. Contrary to popular belief they are both large AND visible. Due to the majority of the erectors sitting below the lats and traps, it has led people to mistakenly say that you can't "see" this muscle group. But by this same logic you would have to conclude that you can't "see" any muscles as they are covered by skin. Both rationales are illogical. Well developed erectors can be seen pushing the lats and traps backwards. They give thick column-like appearances running up the middle of the back. Below you can see underdeveloped erectors on the left vs well developed on the right.

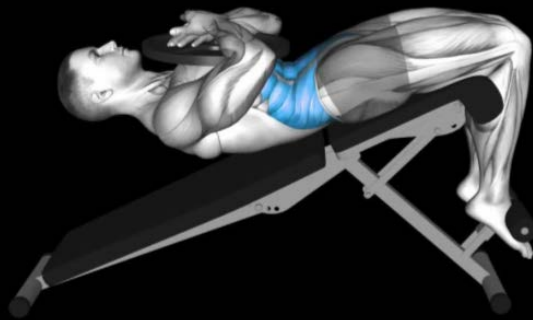
One of the worst things that happened in science based lifting for awhile was the demonization of deadlifts. This led to many people taking heavy hinges entirely out of their program and led to weak and small erectors. Many people will get all the erector development they want so long as they master heavy hip hinges (think SLDLs, RDLs, 45s, etc). However you can do specific exercises to target the erectors if you think those aren't cutting it. My MAIN suggestion is to get these other lifts very strong first BEFORE trying to see if you need hyper specialized work to grow your erectors. It's very rare that someone SLDLing over 500lb for reps is going to need additional erector specialization work. That could look something like this.

But for the most part loading your heavy hinges that work the hamstrings are going to give a robust stimulus to the erectors.

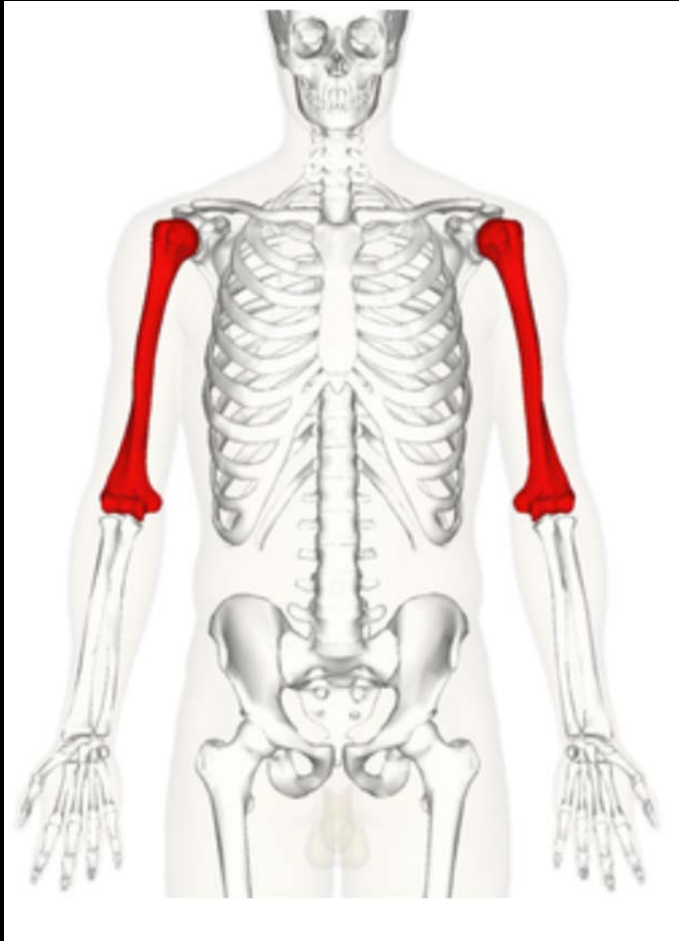
Abdominals



The rectus abdominis (6 pack muscles) is fairly simple. If you are able to load a motion that makes bringing the rib cage towards the pelvis difficult, it will work. There are various ways to do this including cable crunch variations, weighted situp variations, leg raises (that emphasize pelvis movement), abdominal machines, etc. Honestly this work is very boring to talk about and you simply have to find a movement that is stable and loadable that you can do consistently. Treat the abs like any other muscle group (doesn't need excessive volume, reps, frequency like some would have you believe).



And if you have any intent to train obliques as well simply think of it the same way as the rectus abdominis EXCEPT instead of the rib to the pelvis motion being straight up and down, the left side of the rib cage should close the distance towards the right side of your pelvis when working one side of the obliques. Likewise, when working the other side the right side of your rib cage should close the distance towards the left side of your pelvis.

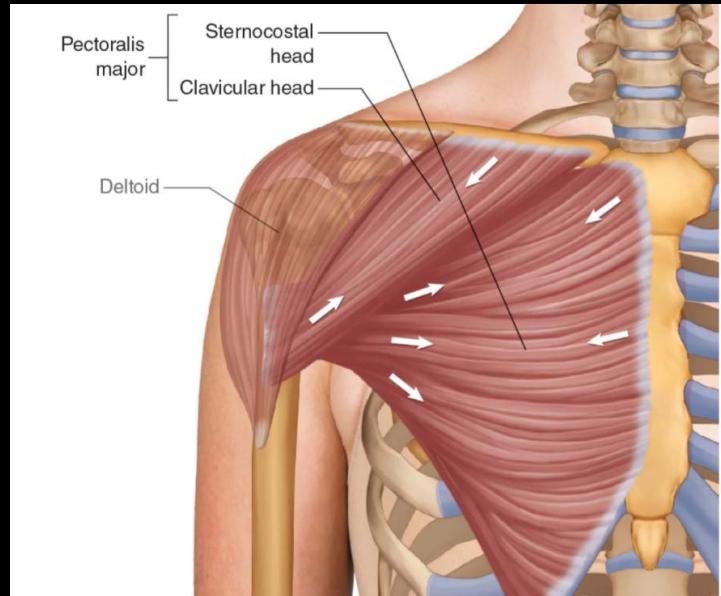


THE HUMERUS

I'm actually writing this after I've completed this manual. Until I began giving descriptions of exercises, I did not realize how important it is to explain what the humerus actually is. It is the bone between your shoulder and your elbow (where your tricep and bicep are located). This will be crucial to know before we move on. Now onto more upper body musculature.

Chest

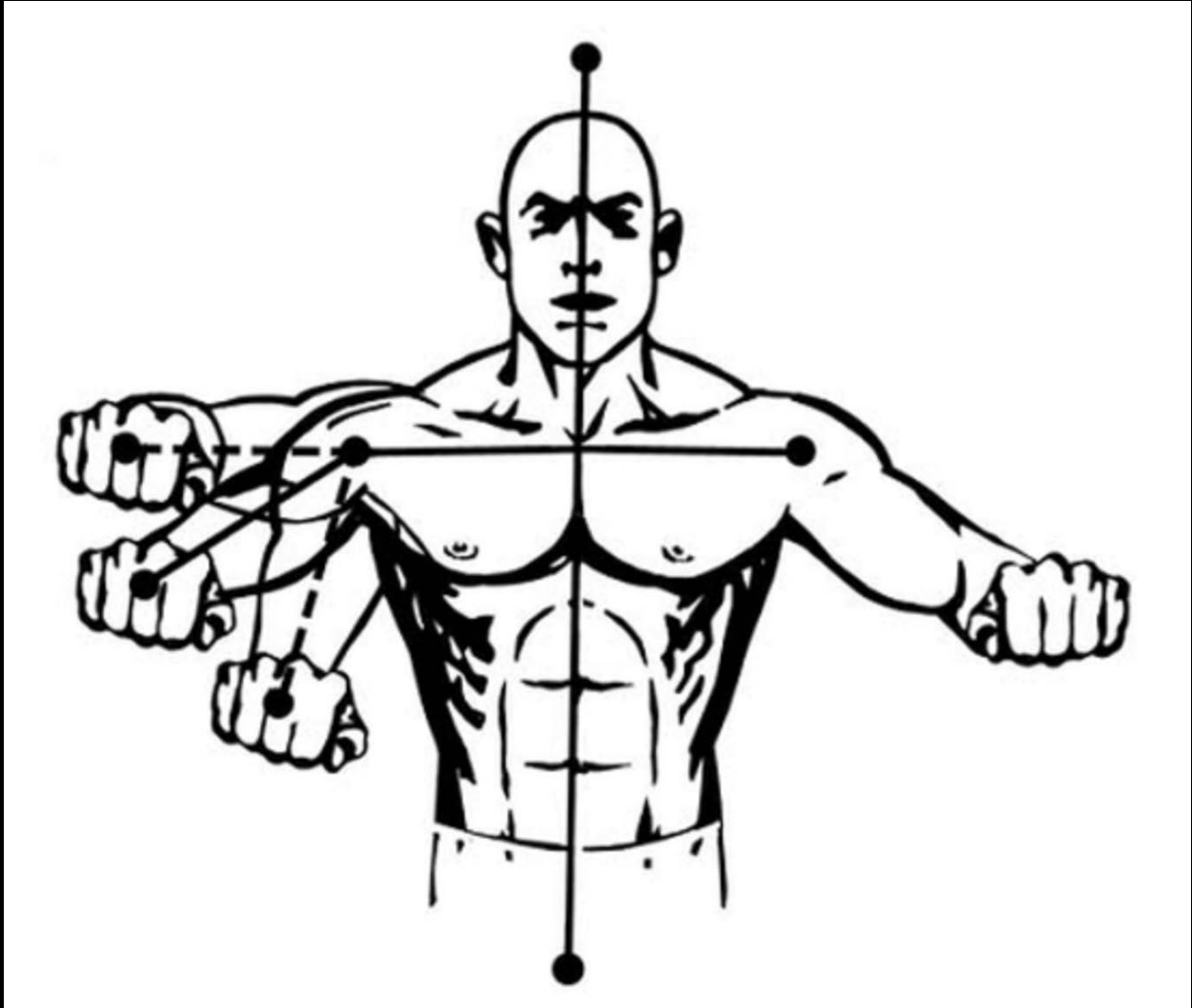
Here's where we get to the interesting stuff. People have often overcomplicated chest training, but if you understand a few basic principles, practical training becomes quite simple. First is that there are two heads. This means two different insertions on the humerus and functions different enough that we definitely want to go out of our way to train each. The primary function of the sternocostal head is horizontal adduction of the humerus (arm moving across the body). This is why flat presses and flat flies work this portion so well. So if you have a solid variation of either a fly or a press in your program, you will cover the majority of the sternocostal head efficiently.



There may be some data to show that you can get slight biasing of the lower chest with different movements as well, however that is incredibly niche and the vast majority of people (even high level bodybuilders) will not need to worry about it in order to get the visual chest growth they are after.

The clavicular head on the other hand does not just play a role in adducting the humerus, but it is also a VERY solid shoulder flexor. This is why we see very high levels of activity in the clavicular head even on front raises that have no adduction component. Due to this (in my opinion) the most effective chest training will have at least two different movements. One of these movements should focus primarily on adduction and the other one should focus primarily on flexion. Keep in mind it doesn't have to be pure flexion. If it has slight adduction that is perfectly fine and perhaps (depending on who you ask) preferable.

The last thing to keep in mind is that even though the path of the arm (tighter to the body) is preferable for the upper pec (clavicular head), we do have some evidence to show that an incline angle for pressing movements does elicit slightly higher activity regardless when compared to flat. People debate the reason for this (a commonly cited argument being leverage) but regardless as to the reason it does seem to be a safe practice when trying to bias the clavicular head to have high amounts of tension in front of the body. On an incline press the hardest portion of the movement is when the humerus is parallel to the ground.

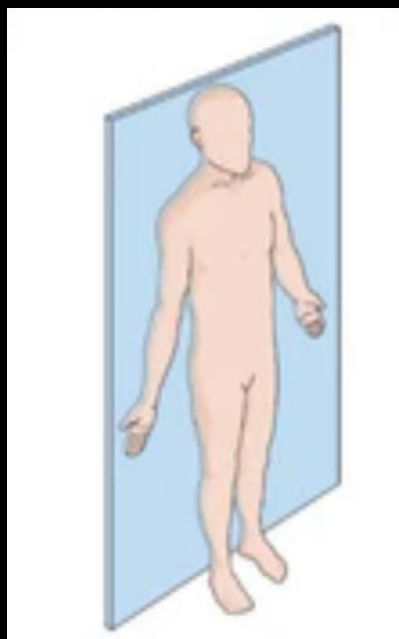
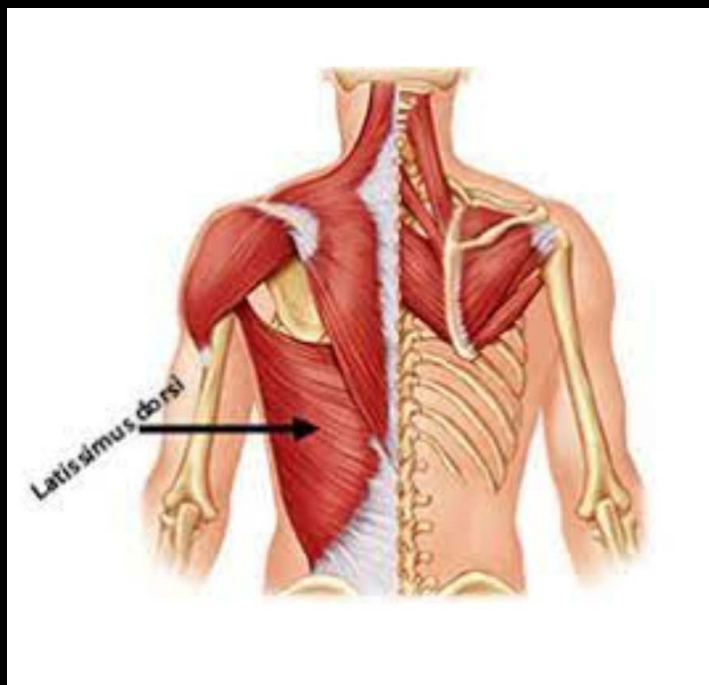


So when looking at this image above it would be important to note that when the arm is at a 90 degree angle to the body the bench should also be flat for more sternocostal biasing. Likewise when the arm is tucked close to the side, the bench should be on an incline for more clavicular biasing. If you want to know how this works for your flys, simply try to match the strength curve that we presented with the presses.

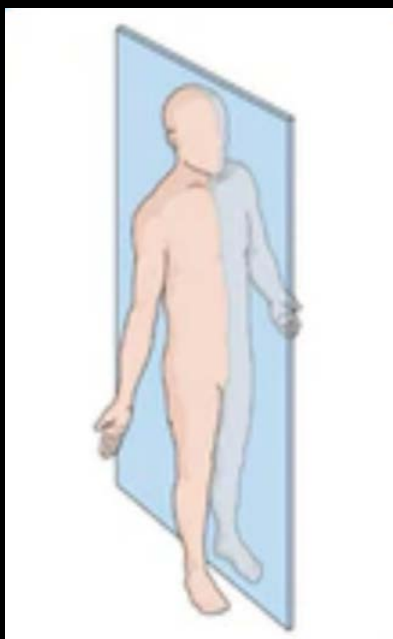
To make a very long story short, whether you choose to do a fly variant or a press, a tucked arm path will make more sense for the clavicular head (upper chest) and a flared arm path will make more sense for the sternocostal head (mid-lower chest).

Lats

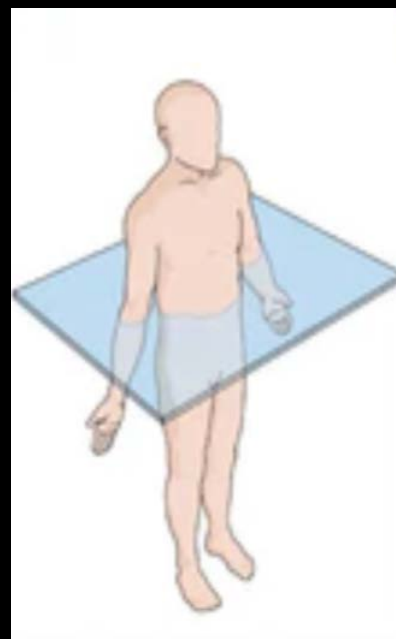
This is a dreadful conversation as there is heavy debate both in how the lats operate and what exercises are best to grow them. Like I've said from the beginning my main goal behind this manual is not to confuse you, but to help you grow. So rather than get into hefty debates over biasing the lower/upper lats, talking through levers, making you work through hyper specific ranges of motion or obfuscating the topic any further by going into every biomechanical argument that people are currently having, I will simply tell you what movements I currently recommend for large lats. And it's simpler than you think.



Frontal (coronal) plane

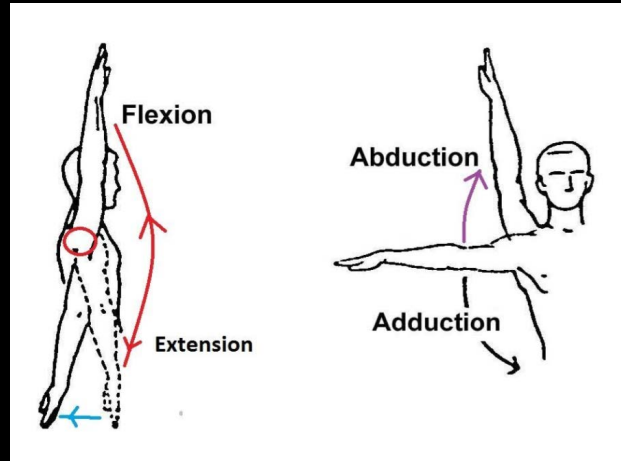


Sagittal plane



Transverse (horizontal) plane

Everything I say from here on out is referring to the movement of the humerus (upper arm). Think of a straight out to the side lateral raise. That would be the humerus moving through the **FRONTAL** plane. Think about a pec deck fly. That would be the humerus moving through the **TRANSVERSE** plane. Think about a front raise. That would be the humerus moving through the **SAGITTAL** plane. Make sure you understand these planes before anything else.

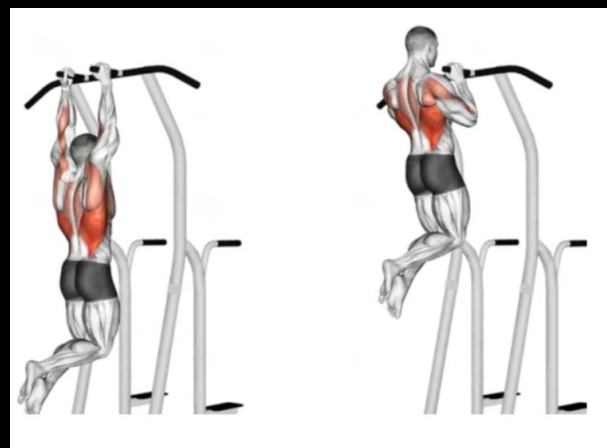
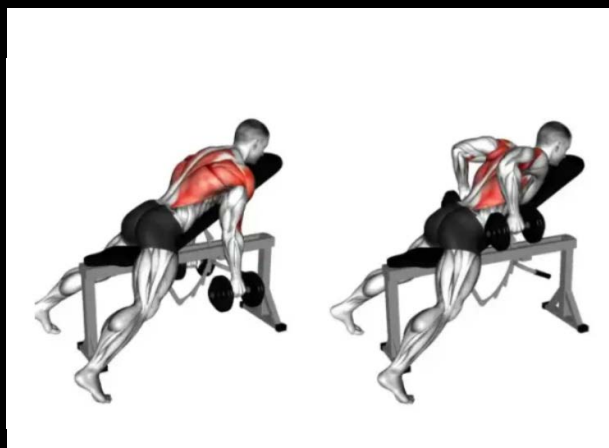
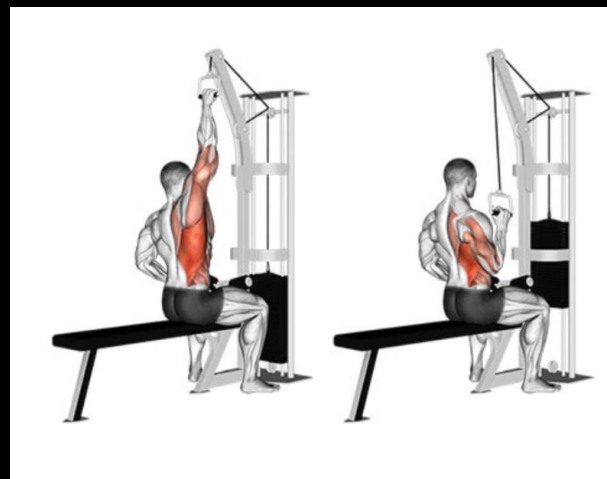


Now that you understand the planes we can begin to discuss the movements you should be looking to perform. The lats play a heavy role in both adduction and shoulder extension. This means that having movements that have resistance opposing these motions will be good for growth. Beginning with the arm overhead and pulling it down towards your side in the **FRONTAL** plane is shoulder adduction.

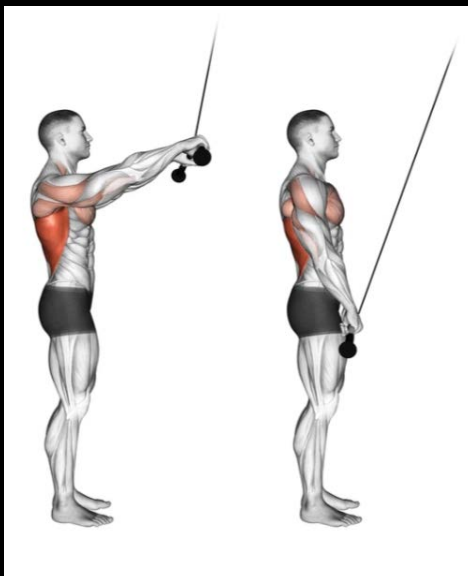
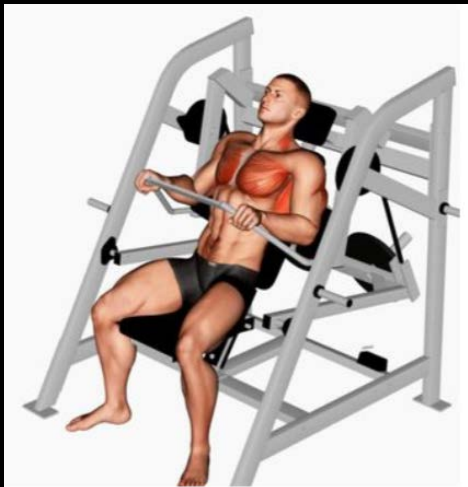
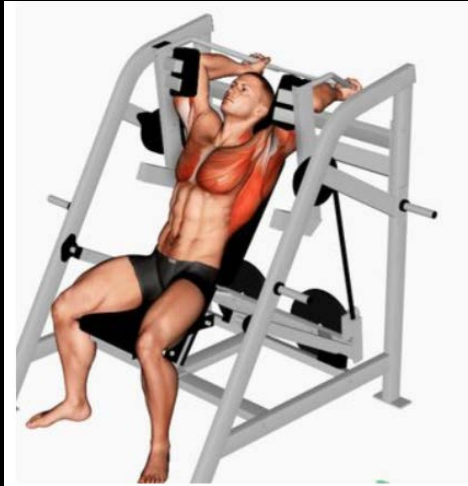


Starting with the arm over head and pulling it down towards your side in the **SAGITTAL** plane is shoulder extension.

It is important with each of these movements to not lean too far back. This allows us to stay closer to the frontal plane and avoid turning it into a pulldown/row hybrid. For shoulder extension people tend to get more creative with the variety of movements they do, but here are many examples.

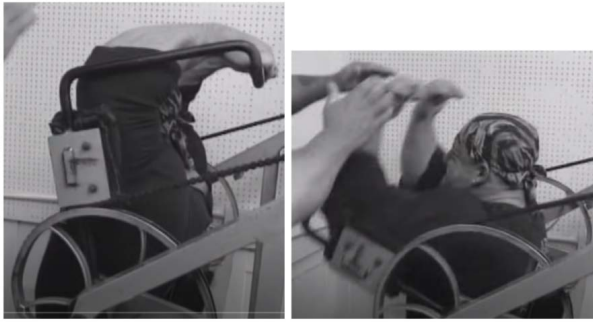


Now I'm going to make something VERY clear. If you progressively overloaded one of the shoulder adduction movements and one of the shoulder extension movements and did ONLY those two exercises for the rest of your life with adequate programming, you would (probably) get the vast majority of lat gains you could possibly want. So if anyone tells you that these are "bad" exercises or that you can't get massive lats with them, this is patently false.



Now that that is out of the way, all of the movements listed to this point have been compound movements (multijoint). You can also train the lats with single joint variations. This is very popular in the sphere currently but it does have its drawbacks. The major drawback is ease of setup. Some have been popular for a long time where others have become popular more recently. Here are some of the exercises you may have seen that can be done in a solid manner for the lats.

If I had to choose between these two exercises, I am most likely to choose the machine pullover. It is more stable, loadable and even has less joints at play due to the force going through a pad at the elbow rather than holding the attachment in your hand. A hotly debated topic currently is how high to allow the humerus to travel as well. Some argue an inability for the lats to produce significant force with the humerus overhead and therefore to only do the bottom two thirds of the motion, while others argue that the lengthening the lats experience overhead will be good for growth.

Top $\frac{1}{3}$ rangeBottom $\frac{2}{3}$ range

PERSONALLY, I find that there is a turnover between the bottom $\frac{2}{3}$ of the range of motion and the top $\frac{1}{3}$ that means strength through the movement is more limited. So REGARDLESS of the arguments people make I train both movements SEPARATELY (Pictured below). The top $\frac{1}{3}$ ends up being its own movement in my programming and the bottom $\frac{2}{3}$ ends up being its own movement in my programming. Here we are getting deep into the weeds though, so if you think these types of nuances are going to make or break your physique, you are sorely mistaken. However you end up deciding to train for yourself, make sure that the movement is stable and loadable and can be progressed over time.

“Flaps”

This deserves its own little section as flaps are a highly contentious topic at the moment, with some saying they are the best movements for lats with others saying they are completely unnecessary and maybe not even a lat movement (argue for it being more teres). As of now I do not see solid argumentation for it being a POOR lat exercise modality. In fact the pullover machine is just a stable version of a sagittal flap. But due to the setup and many of the people currently using flaps, it has received a lot of backlash. This is what a frontal plane flap setup looks like (attachment can go at the elbow or wrist).



I will make a VERY brief summary of the main arguments for and against this exercise. I am only doing this because popular members of the science based lifting community vehemently disagree on this topic.

One side argues that the lats have very good leverage in this position making this a highly efficient method to target the lats and the teres muscle simply comes along for the ride.

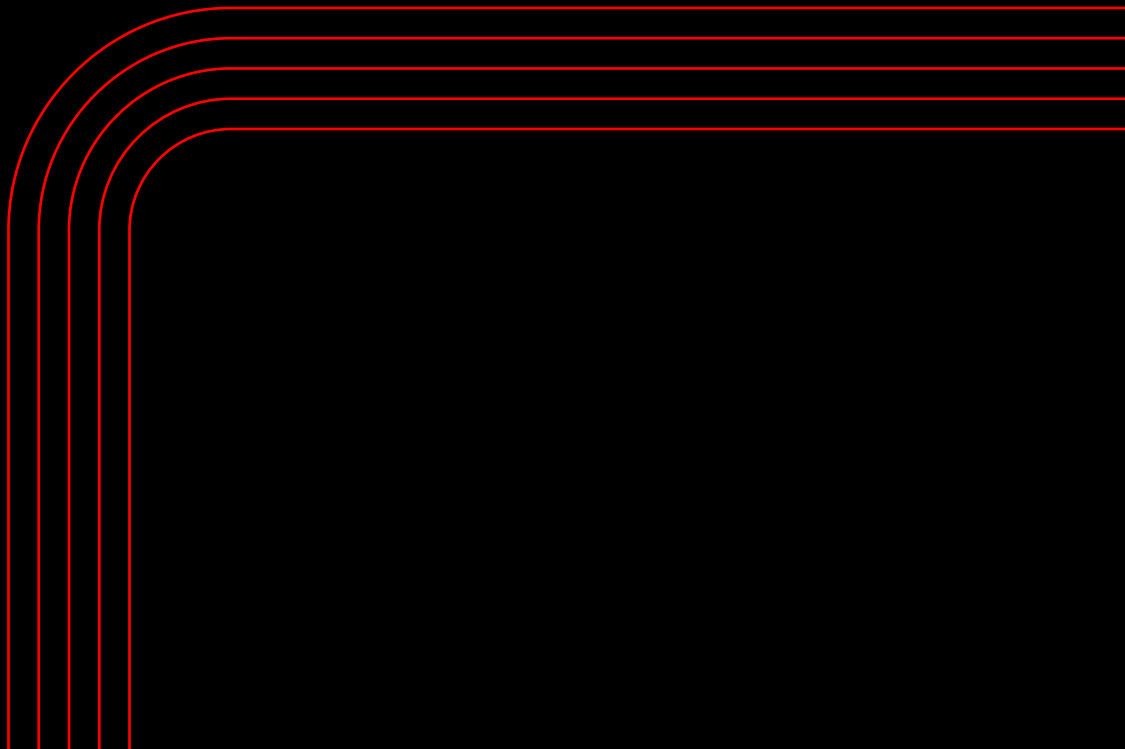
The other side argues that the teres has a high level of advantage in this movement and due to the humerus being limited in traveling upwards (as it would with a pulldown motion) the lats are not lengthening as much nor receiving as much of an advantage as they could.

To make a very long story short NEITHER side argues that flaps will not hit the lats. They simply argue the degree and effectiveness of it. If you want to look a little further into this topic and what people

have argued here are the names of some of the people who have been more involved in these conversations: Keenan Malloy, Yotalks, N1 Coaching, Ben Yanes and many more. I suggest you look into the arguments from all sides and form your own opinion.

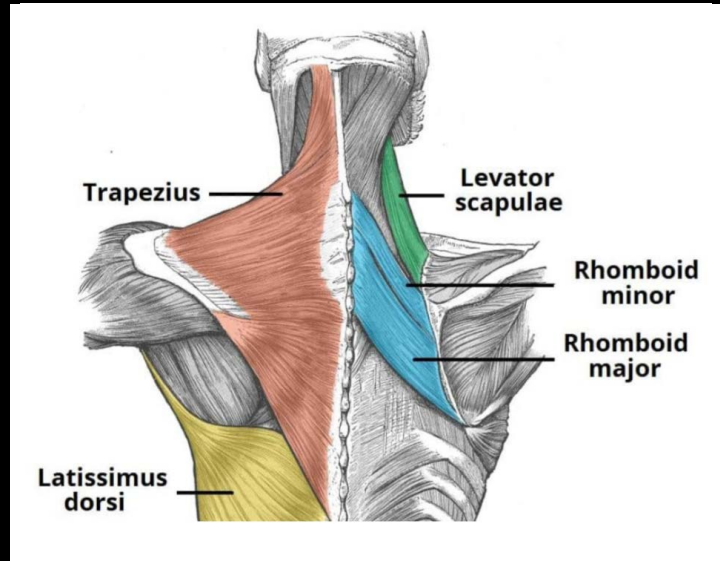
And as I said before (which you'll find to be a common theme with many of these more nuanced conversations) you could have no clue that flaps exist or the conversation surrounding them, have stuck to doing more traditional lat exercises your entire life and got the vast majority if not all of your lat growth. I simply feel obligated to bring up the conversation as it is a hot button issue at the moment (if you're reading this down the road it may be completely settled).

Too long didn't read for the lats; pick a couple stable motions preferably 1 in the frontal (or close to frontal) plane, and one in the sagittal plane and progress those exercises over time for big lats.



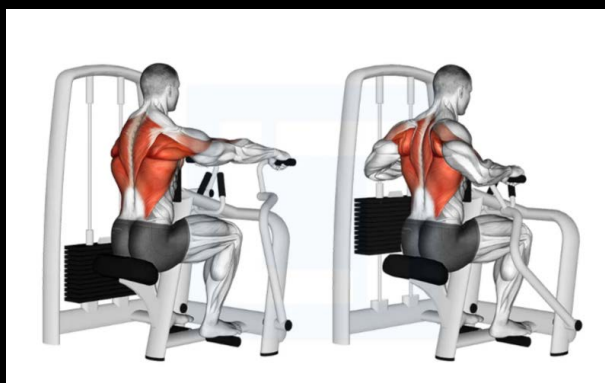
Traps/Rhomboids

The traps and rhomboids play a number of roles but the only real thing you need to be concerned about is how to train them. Due to the varying insertion and origin points, these muscles are fairly complex in terms of all they do, but training them is relatively simple (in my opinion). The upper $\frac{2}{3}$ of the traps should be the primary focus as the lower traps will likely receive a fair amount of indirect stimulus throughout your career. This is not to say that you can't attempt to bias it at times (particularly as an advanced athlete) but the visual differences from this are likely to be much smaller compared to the middle traps, rhomboids, and upper trap development you go out of your way for.



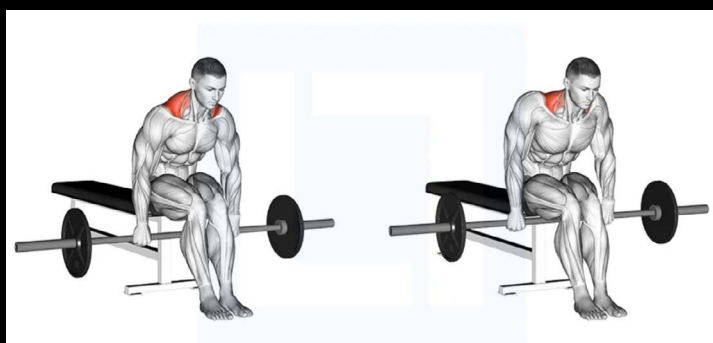
Both rows and shrugs train the traps well. Don't let anyone tell you otherwise. HOWEVER, shrugs are a slightly more direct route to training the traps as it is purely a focus on protraction/retraction or elevation/depression of the scapula. No matter what you decide to program, stability should be a point of emphasis. Chest support is going to allow you to more easily focus on simply working the traps vs coordination/neural demands. Can you still make progress without chest support? Of course. But I highly suggest using it where possible. Having said this, here are some of the exercises I suggest more frequently.





The shirtless man above is clearly not working his traps as resistance is moving in the opposite direction, but I wanted to show you what the protraction and retraction looks like on a horizontal (Kelso) shrug and this was one of the better images I found displaying it. So as long as that motion is being done with the chest supported setups I've shown with rows, you will be getting a solid stimulus.

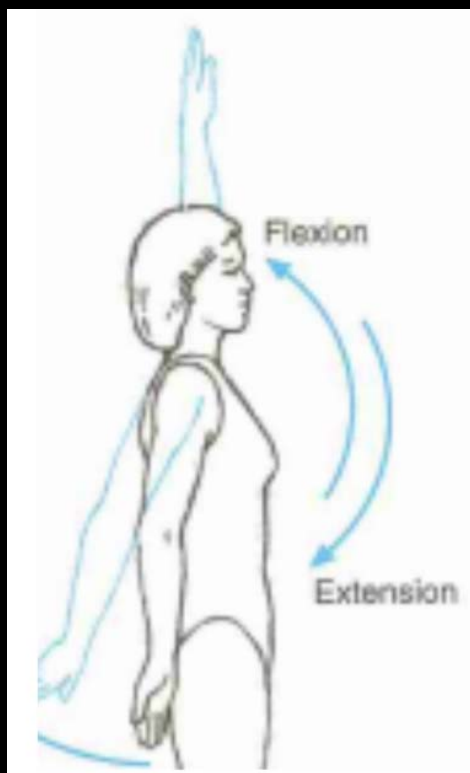
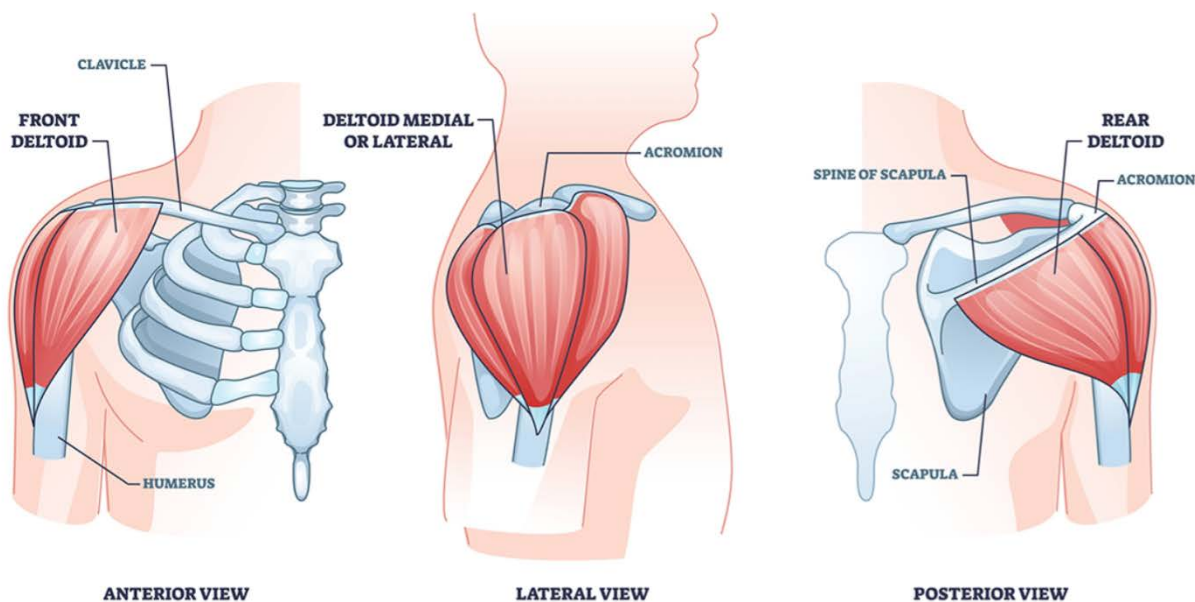
Although a row that has the elbows traveling closer to the sides will still hit the upper traps, they are more likely to involve the lats to a greater degree. My form suggestion (if traps is the goal) is to take a more flared elbow path on the rows. Also, keep in mind that any chest supported row variation can be turned into a kelso shrug by simply protracting and retracting the scapula (pictured to the left).



You can also do shrugs that are slightly more upright. There is debate on these, but I have found that a solid (close to) upright shrug has caused good growth for my mid-upper traps. I have my favorite variation depicted below (however, I prefer them on a smith machine for added stability).

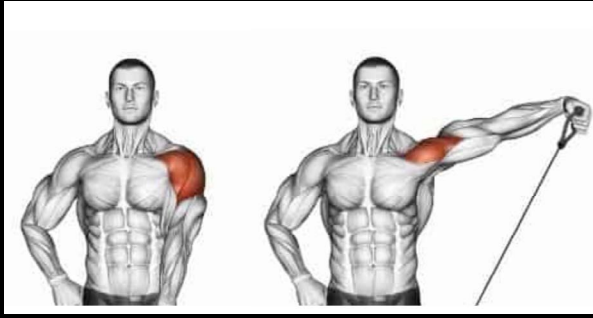
Shoulders

Deltoid Muscles



The shoulders are one of the most unnecessarily overcomplicated muscle groups in my opinion.

The anterior deltoid is a solid shoulder flexor through this entire range and will get solid work through any resisted motion following this path for the humerus (yes, even when the upper chest is heavily involved).

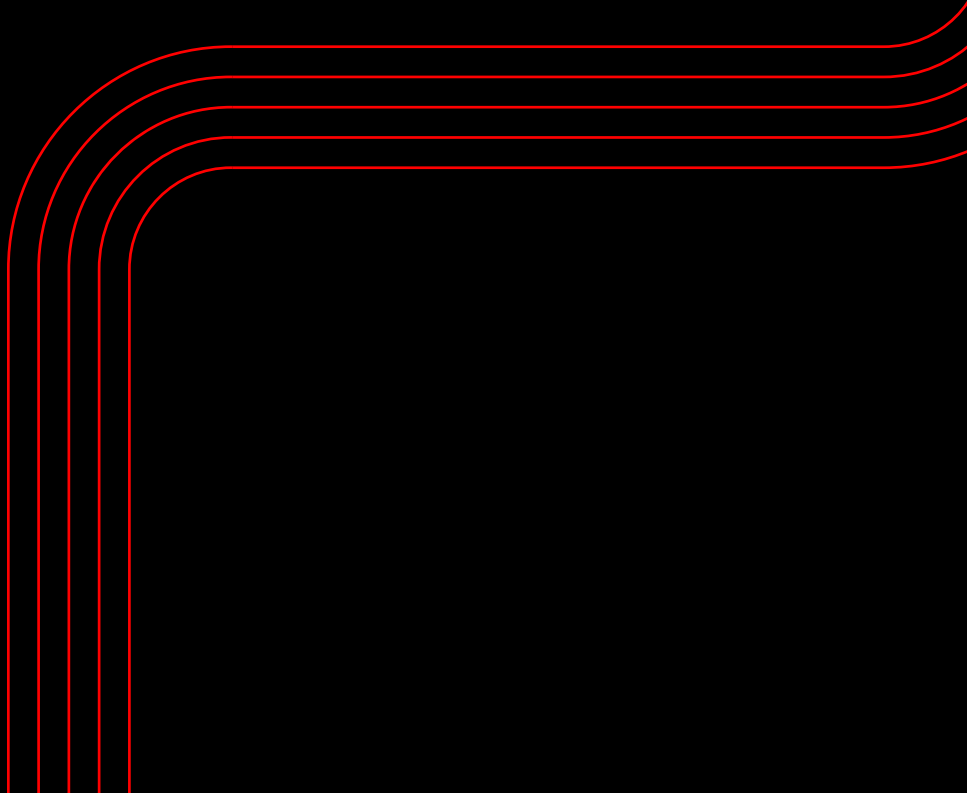


Some solid exercises for the side delts are lateral raise variations (machine, cable, and dumbbell) as well as abducted shoulder presses (and yes this can include behind the neck if you can handle it).

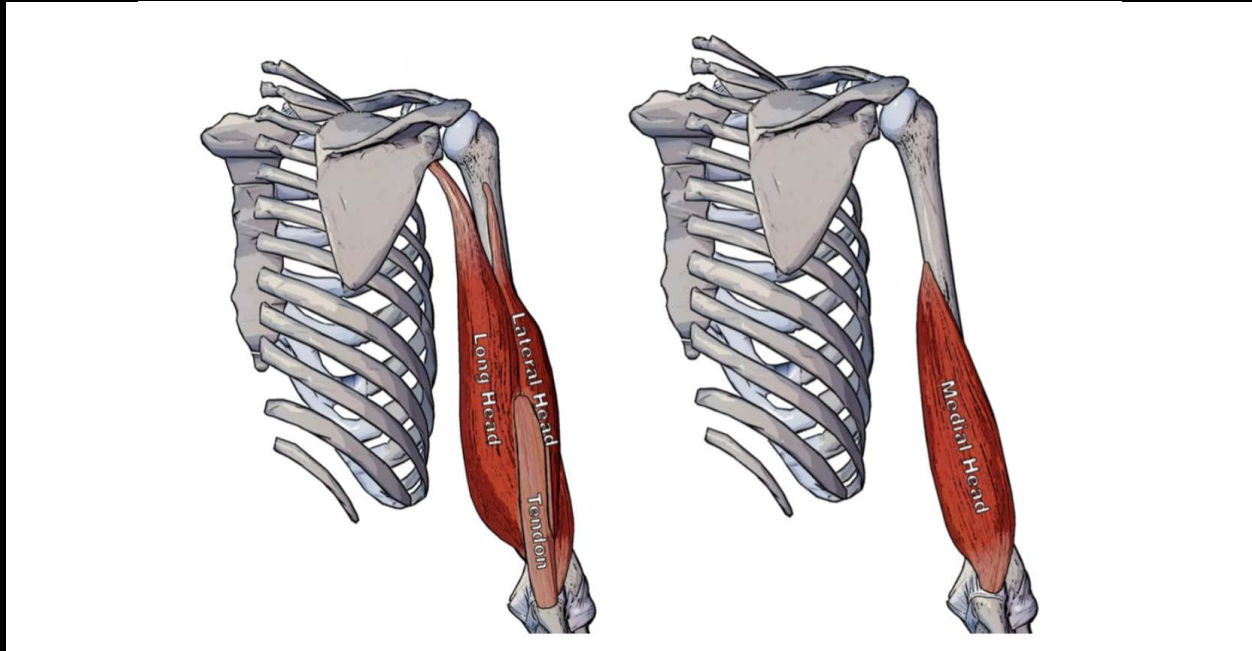
The posterior deltoid contributes to pulling type motions (think shoulder extension, abduction and even external rotation). Due to its heavy role in pulling motions many within the community have moved to say isolating the rear delt is superfluous. I currently tend to disagree, but you can make that judgement for yourself.

Some solid exercises for the rear delts are the reverse pec deck, bent over db raises and rear delt cable extensions.

One thing to note about the shoulders, is the role of each portion is so different (the anterior and posterior even have OPPOSITE functions). So when I hear people talk about volume for the “shoulders” I shake my head. The different portions of the shoulder must be viewed separately when programming. There can be some cross-over (think an abducted shoulder press through a full range of motion which will hit both the front and middle delt hard), but other motions will be entirely separate in terms of stimulus (think front raise vs reverse pec deck).



Triceps

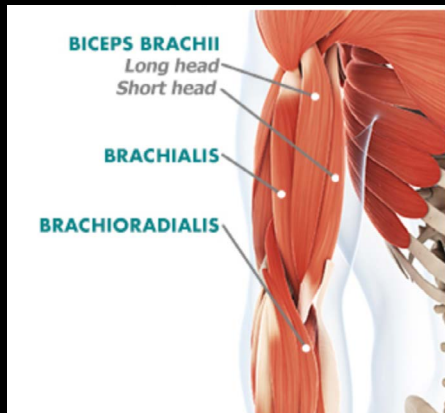


The primary role of the triceps is to extend the elbow, however the longhead also plays a role in shoulder extension. This is because the long head crosses both the elbow and shoulder joint while the lateral and medial head only cross the elbow. This crossing of multiple joints is known as biarticular and it is the same thing we talked about earlier when it comes to the rectus femoris in the quadriceps.

Due to its biarticular nature, even though presses involve elbow extension, we have direct literature to show that presses do not cause meaningful growth for the long head of the triceps. This is why the statement that a chest press works the tris and the chest is only partially true. It works the chest and the medial and lateral head of the tris, but not the longhead. Because of this, even if you were looking to save time with compound movements you would want to include a tricep extension with a fixed humerus in order to get adequate stimulus for the longhead.

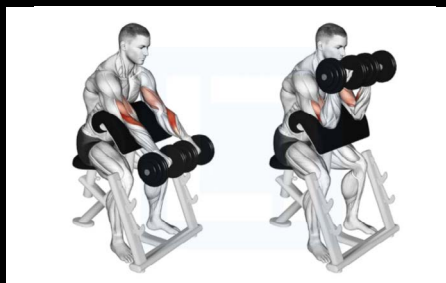
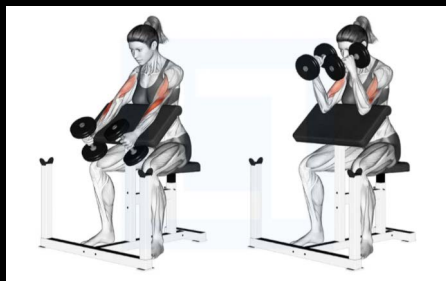
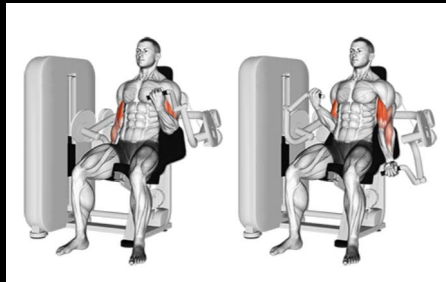
Too long didn't read for the triceps; you will absolutely need an isolated movement to get the majority of the growth out of your triceps as the longhead (the largest head) cannot be worked well with pressing movements.

We can talk about the shoulder extension component for the triceps as well, but honestly I don't think it needs much discussion as the role it will play in long-term triceps growth is likely small (if any). My general recommendation for the triceps is to have at least two movements within your programming. An underhead tricep extension with a fixed humerus and an overhead tricep extension with a fixed humerus are two phenomenal options. However, many people will also want to include a triceps biased press of some sort like a JM. That is totally fine, if not commendable. However, if there are already chest presses in your program having a specific triceps press will be less important.



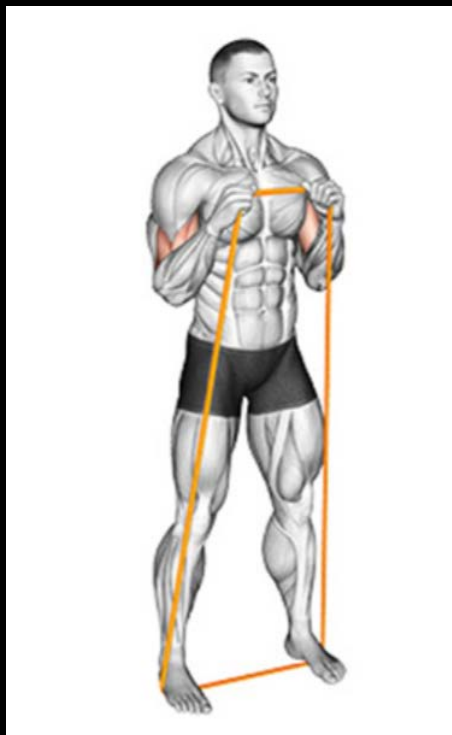
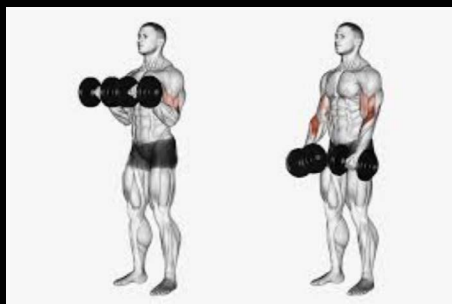
| Biceps (Elbow Flexors)

While Biceps is the main muscle people think of when referring to elbow flexors, we are going to talk about three:



The Brachii: worked very well through elbow flexion, especially when supination of the wrist is involved and possibly greater biasing with a descending resistance profile. However the main thing is to flex the elbow with supination. If you can have elbow support, even better (pictured to the left). There is currently debate surrounding the importance of humerus position. Some argue that a humerus in anatomical position or behind the torso is even better for biceps growth. I cannot confirm that this **MUST** be true, but it is an idea worth exploring in your training (at least for now). Regardless both of these exercise variants are great for brachii hypertrophy

Brachialis: while it is a heavily contentious topic if you can truly “bias” the brachialis or not, it is a large elbow flexor that is worth noting. If we disadvantage the biceps by not supinating the wrist, perhaps the brachialis relatively takes more of the load. I also suggest elbow support for this exercise. One of the exercises I began doing for this was preacher machine hammer curls. I was then challenged further to attempt to disadvantage the biceps even more through full pronation with a descending resistance profile. Essentially a reverse curl that is heaviest at the bottom. Is it possible that this works better? Yes. Do I think that it is going to make a huge difference in your long-term growth? Probably not.



Brachioradialis: The brachioradialis is most definitely going to operate more heavily with a pronated wrist. Reverse curls are the best way to grow this region. It is also possible that an ascending resistance profile (heavier towards the top of the movement) is best for this region. However the most important factor is certainly wrist pronation.

I could go through a wide host of curl variations, but this should give you the general idea as to what you're looking for. The most important thing is wrist position (more supinated→more biceps and more pronated→more brachioradialis). The next is resistance profile with **LIKELY** descending more towards the biceps and ascending more towards the brachioradialis. A good mixture of these variables and you will have some amazing elbow flexors.

Creating a personalized program is NOT something everyone needs to do. Beginner athletes who have not spent enough time in the gym to determine true weakpoints should consider generalized programming before attempting to prioritize certain bodyparts. I want to make this very clear before you accuse me of overcomplicating the process. We discussed 13 muscle group categories above. Here they are listed out:

- Calves
- Quads
- Hamstrings
- Adductors
- Glutes
- Abs
- Erectors
- Chest
- Lats
- Traps
- Deltoids
- Biceps
- Triceps

Body parts that receive a 3 should always be at the end of your sessions if you are looking to develop a more balanced physique. Body parts that receive a 1 should be at the beginning of your sessions, and body parts that receive a 2 should be in the middle.

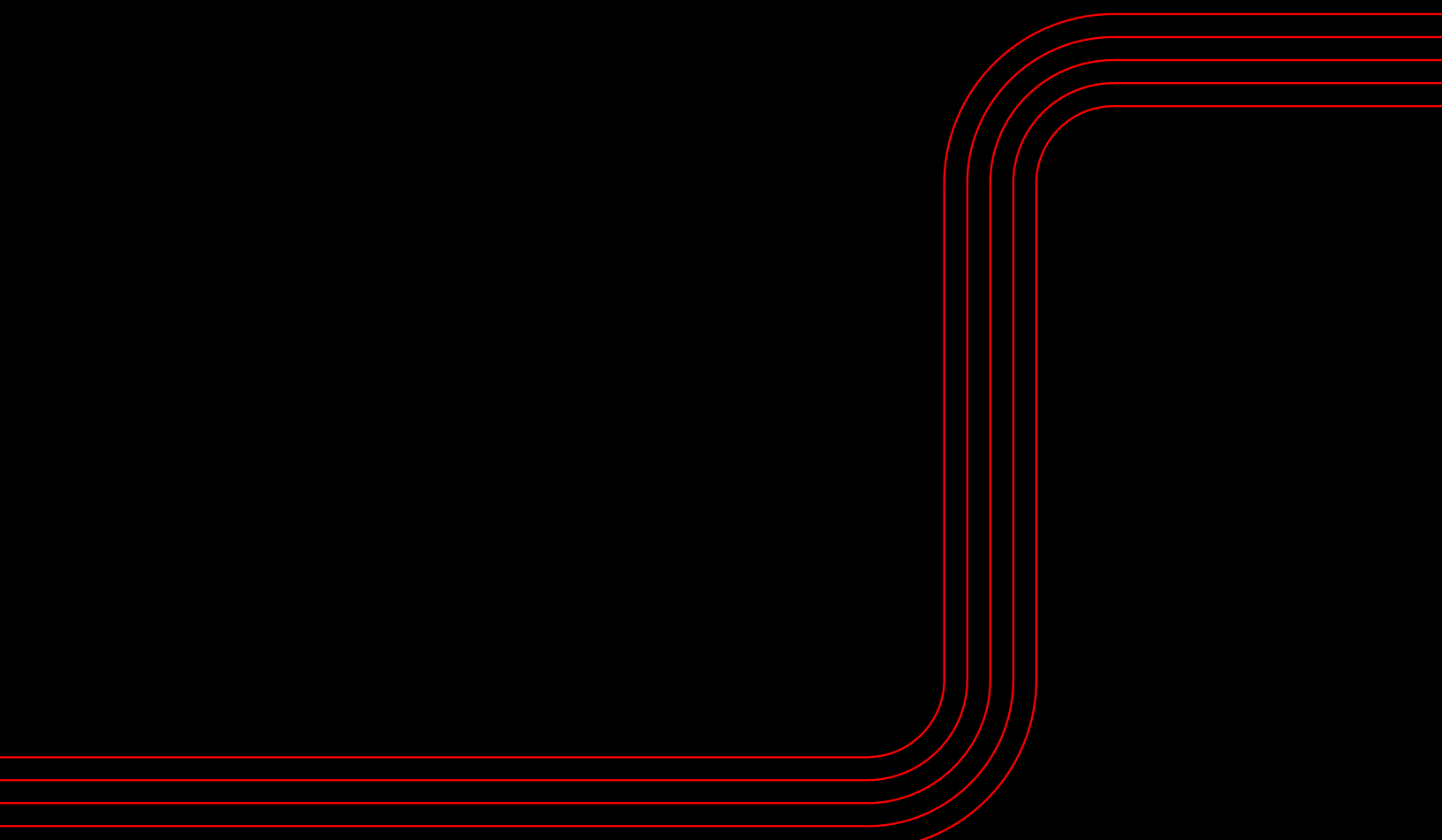
Calves-2
Quads-2
Hamstrings-1
Adductors-3
Glutes-1

- Seated Hamstring curls
- Hip Thrusts
- SLDLs
- Leg Extensions
- Seated Straight Leg Calf Raises
- Squat Pattern
- Adductor Machine

Now some men don't care too heavily about growing their glutes. So even if it is a weak point, maybe you exclude it from the discussion entirely (I disagree with the choice, but it is YOUR program). There are no specific rules, only general guidelines if you are trying to maximize growth and develop a balanced physique.

With all the principles we have talked about you SHOULD be able to program your own split and make it quite good. But just in case you're still struggling with some of the concepts, I have provided some sample splits using some of the more popular general layouts. There is a TON of individual variety you can implement, so understand that EVERYTHING you see is guidelines. You can tailor exact set/rep numbers to your preferences. You can take more or less rest days. There are NO rules, but if you ensure continued progress on these movements, you WILL grow. It is that simple.

Thank you for all the support and I truly hope this makes your lifting journey easier. For the people new to lifting I hope this wasn't too complicated to understand, and for the super nerds, I hope there was still some value you could garner from this. Regardless, I am very supportive of everyone's journey and make it a point to interact with my audience regularly, whether that's through video/comment replies (not DMs) or answering questions on lives. If you are persistent in seeking me, I should be able to clarify anything you need. As always I love you guys, appreciate you guys, TNF out.



SAMPLE PROGRAMS

JOEL TWINEM

FULLBODY

REPEATED 2-3.5X/ WEEK

EXERCISE	REPS
**Pec Deck or Flat Press	1-2x5-10
Lat Pulldown/Stable Pullup	2x5-10
Lateral Raise Cable or DB	2x5-10
**Incline Elbow Tucked Chest Press	1-2x5-10
Chest Supported Row or Kelso Shrug	2x5-10
**Preacher Curl	1-2x5-10
Cable Tricep Extension	2x5-10
Leg Extension	1-2x5-10
Seated Hamstring Curl	1-2x5-10
Adductor Machine	1-2x5-10
Calf Raise	2-3x5-10
Hip Thrust	1-2x5-10
**Hack Squat	1-2x5-7
**SLDL	1-2x5-7

UPPER/LOWER

REPEAT AS DESIRED BUT MINIMUM 2X/WEEK

EXERCISE	REPS
UPPER	Lat pulldown 2-4x5-10
	**Pec Deck or Flat Press 1-3x5-10
	Chest Supported Row or Kelso Shrug 2-4x5-10
	**Incline Elbow Tucked Chest Press 1-3x5-10
	Lateral Raise DB or Cable 2-4x5-10
	Underhead Tricep Extension 1-3x5-10
	**Overhead Tricep Extension 1-3x5-10
	Preacher Curl 1-3x5-10
LOWER	**Reverse Pec Deck 1-2x5-10
	Leg Extension 2-3x5-10
	Seated Hamstring Curl 2-3x5-10
	Adductor Machine 2-3x5-10
	Calf Raise 2-4x5-10
	Hip Thrust 2-3x5-10
	**Hack Squat 1-2x5-7
	**SLDL 1-2x5-7

NOTES

- 1** Total daily sets should be dependent on what YOU believe you can recover from reasonably
- 2** If you ascribe to the belief you can only recover from less than 15 total sets per day, be more selective with your exercises and set numbers
- 3** The listed sets are guidelines not rules
- 4** Every listed exercise should be done at least 1x/week (preferably 2)
- 5** If every exercise can be performed with adequate intensity and progression they CAN be all done 3x/week
- 6** The starred exercises are the suggested adjustments from 3 to 2x/1x week if you are looking to lessen total volume but are programming 2-3x/week in general
- 7** Hack Squats and SLDLs overlap the most with other muscle groups worked
- 8** Exercise order is not set and should be based on your focuses or weak points (weak points earlier in session)
- 9** **IMPORTANT:** more does not always mean better so if you assume doing highest possible number of sets is best, you could get burned
- 10** Flys and presses can be interchanged to an extent (just keep in mind the extra tricep fatigue with presses)
- 11** It is important to note for ALL exercises that the main goal is to hit the target tissue with stability and the right resistance profile

- 12** If you want to swap out an exercise you always can, just make sure to not be swapping constantly
 - 13** Once you find an exercise you like that accomplishes a similar purpose to the exercise listed in the program, stick with it for a solid amount of time (months to years not weeks)
 - 14** You WILL have intelligent people tell you that these programs are overcomplicated and under complicated. You need to decide for yourself where you fall on this spectrum and there isn't necessarily a right or wrong answer
 - 15** I ENCOURAGE you to think through your stances, but do NOT become an exercise or program hopper constantly switching. This WILL hinder longterm gains.
 - 16** Exercise order can be rearranged based on what YOU specifically are trying to prioritize
 - 17** Once you know what plan you are going to use, program it into a tracking app or digital or physical notepad. If you don't have a subscription to one there are apps you can use for free (like Boostcamp) so you can continue to log progress week after week
 - 18** Every few weeks do video form checks to make sure progress is happening WITH the same technique and you aren't changing it in order to force "progress". If you find that you are, simply reduce the weight until your form gets back on track.
 - 19** If you think there are too many exercises in a day (which is a perfectly reasonable stance), pick the ones that cover the most muscle groups and stick with them. For example, if you need to remove two exercises from the "Pull Day" it makes more sense to remove one of the lat exercises and one of the trap exercises than both lat exercises.
- 